# Classical training of conditional instantaneous quantum polynomial-time Born machines

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Instantaneous quantum polynomial-time (IQP) Born machines allow classical estimation of Pauli-$Z$ expectation values required for Gaussian-kernel maximum mean discrepancy (MMD) training, decoupling optimization from quantum hardware. However, their standard formulation learns a single distribution. We introduce a classically trainable conditional framework that encodes conditions in input states and jointly trains one shared IQP unitary. The trained unitary remains fixed when the input state encodes new conditions, including those absent from training. Notably, the encoder need not have IQP form, provided the reference state entering the diagonal IQP layer supports efficient classical sampling and amplitude-ratio evaluation. By combining amplitude-ratio estimation of these expectation values with the MMD formulation, we derive unbiased classical estimators of the conditional objective and its gradients. Furthermore, the framework supports input states that impose exact output symmetries in both conditional and unconditional models. In sixteen-qubit numerical experiments, we validate conditional generation through translations of Gaussian-shaped profiles. A correlated phase encoder then enables joint center and width control at combinations absent from training. For the Curie–Weiss model, our shared conditional model reproduces the temperature-dependent change from unimodal to bimodal magnetization profiles, including at temperatures absent from training. Finally, classical training reduces held-out sampled $\mathrm{MMD}^2$ in models with up to 2048 qubits.

## I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ within a single model. They represent how a distribution depends on a condition, such as temperature in physics or a class label in image generation. Learning across conditions aims to capture structure common to the family and support generation at conditions absent from training. Classical examples include conditional generative adversarial networks, variational autoencoders and diffusion models [1–3]. Quantum circuit Born machines generate samples by measuring parameterized quantum circuits in the computational basis. They also support conditioning through input-state preparation, condition registers or postselection [4–6].

Instantaneous quantum polynomial-time (IQP) circuits offer a useful separation between output sampling and expectation-value estimation. Specifically, certain IQP circuit families are classically hard to sample from under standard complexity assumptions [7, 8]. Nevertheless, classical algorithms can efficiently estimate expectation values of Pauli-$Z$ products for these same circuits to additive accuracy [9]. This distinction enables classical training of IQP Born machines because the Gaussian-kernel squared maximum mean discrepancy (MMD) objective can be expressed through these expectation values [10, 11]. Consequently, classical algorithms can estimate both the objective and its gradients, allowing optimization without quantum hardware [12].

Building on classical IQP training, recent methods introduce control by adapting parts of a generator. For example, ConQuER fits an additional controller to data filtered by a target condition while keeping a pretrained generator fixed [13]. Similarly, Liu *et al.* retain a shared circuit core, fit a small parameter block separately at selected conditions, and interpolate the fitted parameters to generate at unseen conditions [14]. Their subsequent work uses physics-informed initialization to improve condition-specific latent adaptation [15]. These methods therefore introduce condition dependence through fitted controller or latent parameters. In our framework, by contrast, a prescribed encoder maps conditions to input states, and we train one shared IQP unitary jointly across conditions. Once trained, this unitary remains fixed, so new conditions enter directly through the encoder without fitting condition-specific parameters or interpolating a learned parameter trajectory.

We introduce a classically trainable conditional IQP framework in which an encoder prepares $|\phi(c)\rangle$ and a shared learned unitary defines $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. Notably, the encoder need not have strict IQP form, provided its reference state $|\psi_c\rangle = H^{\otimes n}|\phi(c)\rangle$ supports efficient classical sampling and amplitude-ratio evaluation. These operations let us apply established sparse-observable estimation methods to obtain unbiased estimates of conditional Pauli-$Z$ expectation values and their phase derivatives [9, 16]. Combining these esti-

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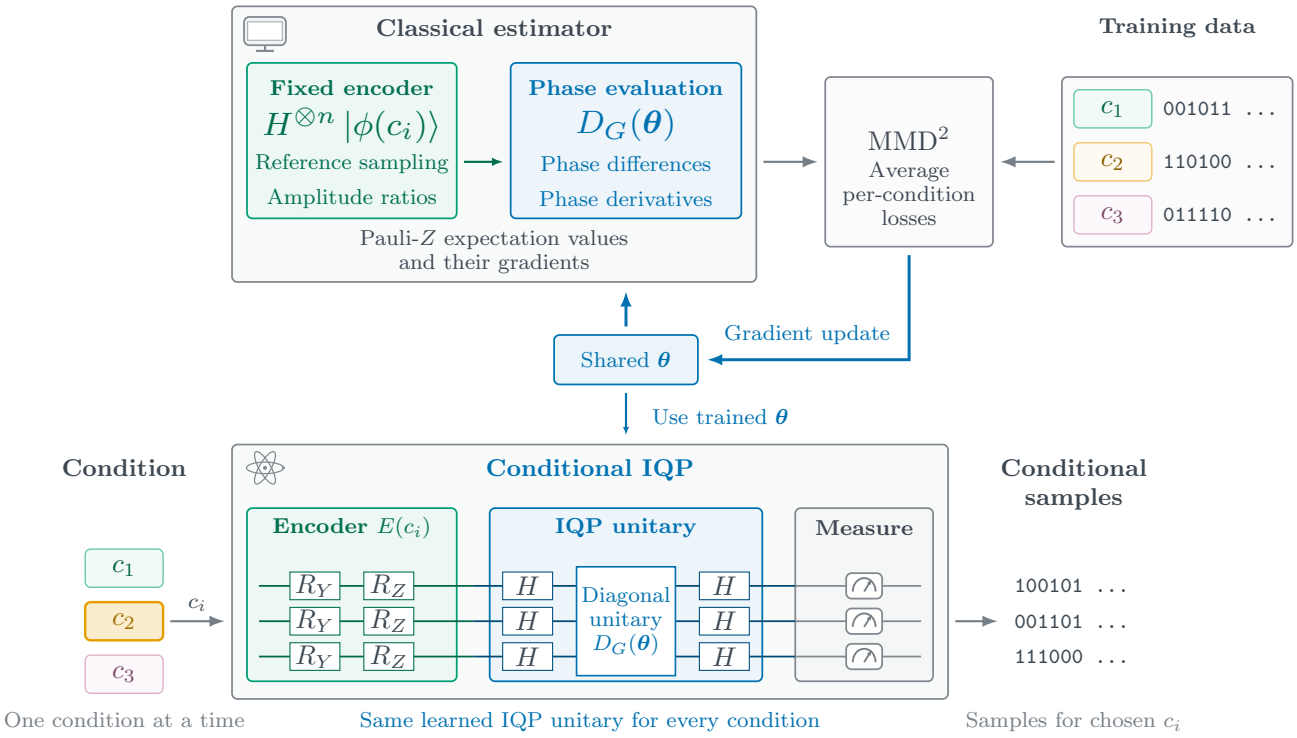



FIG. 1. Proposed framework for classical training and quantum sampling of conditional instantaneous quantum polynomial-time (IQP) Born machines. An encoder maps each condition $c_i$ to an input state $|\phi(c_i)\rangle$. The local rotations illustrate a product-state encoder; the framework also admits correlated encoders that satisfy the reference-state access requirements. During classical training, samples and amplitude ratios from the reference state $H^{\otimes n}|\phi(c_i)\rangle$, together with phase differences induced by the IQP layer, yield estimates of conditional Pauli-$Z$ moments and their gradients. We use these estimates to optimize one shared parameter vector $\boldsymbol{\theta}$ with a squared maximum mean discrepancy (MMD) objective averaged across training conditions. During quantum sampling, a chosen condition $c_i$ determines the input state. Applying the trained IQP unitary $U_{\boldsymbol{\theta}}$ and measuring in the computational basis produces samples from the learned conditional distribution $q_{\boldsymbol{\theta}}(\cdot \mid c_i)$.

mators with the moment formulation of squared MMD then yields unbiased estimates of the conditional training objective and its gradients [11, 12]. Furthermore, we construct correlated reference states whose parity constraints enforce prescribed output bit-flip symmetries in conditional and unconditional models. These states retain efficient classical sampling and amplitude-ratio evaluation. Figure 1 summarizes the framework.

We validate the framework through sixteen-qubit numerical experiments. First, we translate Gaussian-shaped profiles to demonstrate the basic conditioning mechanism. We then introduce a second phase-encoding coordinate to control profile width and generate at combinations absent from training. Both experiments use independently parameterized lattice layers and share an exact binomial center fixed by construction. To test larger changes in distributional shape, we use the Curie–Weiss model, where one shared conditional model reproduces the change from unimodal to bimodal magnetization profiles as temperature varies, including at temperatures absent from training. As a worked example of the symmetry construction, we also give a signed-graph encoder that imposes several independent output symmetries. Finally, we establish computational feasibility through classical training runs with up to 2048 qubits.

Sections II and III describe the model and classical training. Section IV reports the experiments, and Section V discusses input-induced symmetries. Sections VI and VII discuss the findings and conclude.

## II. CONDITIONAL IQP MODEL

We separate condition encoding from the learned unitary transformation. An encoder maps each condition $c \in \mathcal{C}$ to an input state $|\phi(c)\rangle$, and we apply the same parameterized IQP unitary $U_{\boldsymbol{\theta}}$ to every encoded state. For a target family $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over $n$-bit strings, we define the model distribution as

$$q_{\boldsymbol{\theta}}(x \mid c) = |\langle x| U_{\boldsymbol{\theta}} |\phi(c)\rangle|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

Throughout this work, we use "IQP unitary" to denote only the learned block $U_{\boldsymbol{\theta}}$. The encoder need not have IQP form and may include non-diagonal operations, so

the complete conditional circuit need not belong to the strict IQP circuit class.

We parameterize the shared IQP unitary by an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator $Z_i$, and each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. We write the unitary as

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left( i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j \right). \tag{3}$$

All generators in $D_G$ are diagonal in the computational basis and commute. In the numerical experiments, we use only one- and two-body generators. However, this formulation extends naturally to $k$-body interactions by adding the corresponding Pauli-$Z$ products to the exponent.

The initial Hadamard layer maps the encoded input to the reference state $|\psi_c\rangle = H^{\otimes n} |\phi(c)\rangle$, immediately before the diagonal IQP layer. We assume that each encoded input is normalized and independent of the learned parameters $\boldsymbol{\theta}$, so the reference state has the same properties. We train the model using samples and amplitude ratios from this reference state. Its computational-basis amplitudes are

$$\psi_c(z) = \langle z | \psi_c \rangle = \langle z | H^{\otimes n} | \phi(c) \rangle , \tag{4}$$

The squared amplitudes $|\psi_c(z)|^2$ give the reference sampling distribution for condition $c$. Our requirements concern classical access to this state rather than the encoder's gate structure. For the relevant conditions $c$ and bit masks $a$, we require two efficient classical operations:

1. Sample $z \sim |\psi_c(z)|^2$.
2. Evaluate the amplitude ratio

$$R_c(z, a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \tag{5}$$

for sampled $z$, where $\oplus$ denotes bitwise addition modulo two.

We call these two operations the sampling-and-ratio interface.

Throughout, we assume polynomial-size descriptions of the encoder and conditions and polynomially many IQP generators. Efficient access requires polynomial computational cost in $n$, these description lengths and the inverse accuracy needed for estimation. Our analytical guarantees assume exact sampling and arithmetic. Finite-precision implementations must also control sampling and numerical errors.

We estimate Pauli-$Z$ moments by combining reference amplitude ratios with phase factors from the IQP layer. For each sampled string $z$, the ratio $R_c(z, a)$ compares the reference amplitudes of $z$ and the paired string $z \oplus a$. The IQP layer contributes a complex phase factor determined by the phase difference between these strings. We multiply the amplitude ratio by this phase factor and average the real part over reference samples to estimate the corresponding Pauli-$Z$ moment [9].

We evaluate amplitude ratios only on the support of $|\psi_c|^2$, because reference sampling never selects strings with $\psi_c(z) = 0$. Computationally tractable reference states in the sense of Van den Nest admit efficient computational-basis sampling and amplitude evaluation [9]. These states provide a sufficient realization of the sampling-and-ratio interface when amplitude evaluation supplies the precision needed to compute the required ratios efficiently.

The simplest case is a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^{n} |\phi_i(c)\rangle$. Its reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b | H | \phi_i(c) \rangle$. We can therefore generate reference samples by drawing each bit independently, and the amplitude ratio is

$$R_c(z, a) = \prod_{i : a_i = 1} \frac{\psi_{c,i}(1 - z_i)}{\psi_{c,i}(z_i)}. \tag{6}$$

When the local amplitudes have constant arithmetic evaluation cost, both operations require at most $O(n)$ local operations.

Beyond this product-state example, Section V constructs correlated inputs that satisfy the same access requirements and impose exact output symmetries.

The encoder also constrains how the model's output probabilities depend on the condition $c$. For rotation encodings with angles linear in $c$, the encoding generators determine the accessible Fourier frequencies, while the learned parameters control their coefficients within the constraints of the circuit architecture [17]. Multidimensional Fourier analyses further describe how the encoding and circuit architecture constrain the model's expressive capacity [18]. Caro et al. also derive encoding-dependent generalization bounds for parameterized quantum circuits [19]. We therefore distinguish encoder expressivity and generalization from the requirements for efficient classical estimation.

Efficient classical estimation of the observables used for training does not imply efficient classical sampling from $q_{\boldsymbol{\theta}}(\cdot \mid c)$. Existing IQP sampling-hardness results apply to specific circuit families and do not establish hardness for arbitrary condition-dependent input states or trained parameter values [7, 20]. To generate conditional samples on a quantum computer, we also need efficient circuits that prepare $|\phi(c)\rangle$ and implement $U_{\boldsymbol{\theta}}$. The classical sampling-and-ratio interface alone does not guarantee such circuits.

### A. Conditional Pauli estimator and gradients

We express the conditional MMD objective in Section III in terms of Pauli-$Z$ moments. For a bit mask $a \in \{0,1\}^n$, we define $Z_a = \bigotimes_{i=1}^n Z_i^{a_i}$ and denote its conditional model expectation by $\mu_q(a \mid c)$. We use the sampling-and-ratio interface to derive estimators for this moment and its derivatives with respect to the learned phases.

We specialize established classical estimation methods to conditional IQP reference states. Van den Nest established efficient sparse-observable estimation for computationally tractable states [9], and Havlíček developed estimation using sampling and amplitude-ratio access [16]. Recio-Armengol *et al.* applied classical Pauli-moment estimation to MMD training of IQP Born machines [12]. Here, we apply these methods to prescribed condition-dependent reference states, enabling joint training of one shared IQP unitary across conditions.

We denote the diagonal IQP layer by $D_{\boldsymbol{\theta}}$ and write its action as

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle , \tag{7}$$

with

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \tag{8}$$

where $P$ is the number of generators and $g_j \in \{0,1\}^n$ specifies the support of the $j$th Pauli-$Z$ generator. We evaluate the dot product $g_j \cdot z$ modulo two. We define the phase difference associated with the mask $a$ as

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \tag{9}$$

The final Hadamard layer transforms $Z_a$ into $X_a = \bigotimes_i X_i^{a_i}$. Consequently, the conditional Pauli moment satisfies

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c | \, D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}} \, |\psi_c\rangle \\ &= \sum_z \psi_c(z)^* \psi_c(z \oplus a) \\ &\quad \times e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}. \end{aligned} \tag{10}$$

Terms with $\psi_c(z) = 0$ vanish. On the reference sampling support, we can write

$$\psi_c(z)^* \psi_c(z \oplus a) = |\psi_c(z)|^2 R_c(z,a). \tag{11}$$

The sum therefore becomes an expectation over reference samples, with the amplitude ratio multiplied by the IQP phase factor. Since the Pauli moment is real, we take the real part of this product.

For a reference sample $z \sim |\psi_c|^2$, we define

$$\begin{aligned} W_{c,a}(z;\boldsymbol{\theta}) &= R_c(z,a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z;\boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}). \end{aligned} \tag{12}$$

We define the derivative of the phase difference with respect to $\theta_j$ as

$$\begin{aligned} d_{j,a}(z) &= \partial_{\theta_j} \Delta_a f_{\boldsymbol{\theta}}(z) \\ &= \chi_{g_j}(z \oplus a) - \chi_{g_j}(z) \in \{-2, 0, 2\}. \end{aligned} \tag{13}$$

Fix a condition $c$ and a mask $a$. We assume that the reference state is normalized, independent of $\boldsymbol{\theta}$, and supports the sampling-and-ratio interface. For each reference sample, the phase-gradient estimator is

$$Y'_{j,c,a}(z;\boldsymbol{\theta}) = \mathrm{Re}[i \, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta})] . \tag{14}$$

For each fixed condition, define

$$\begin{aligned} A_a &= D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}}, \\ B_{ja} &= \partial_{\theta_j} A_a = i[A_a, Z_{g_j}], \end{aligned} \tag{15}$$

Both operators are Hermitian and have at most one nonzero entry per row. Their local sparse-observable estimators have real parts $Y_{c,a}$ and $Y'_{j,c,a}$, respectively. Since the reference state is independent of $\boldsymbol{\theta}$, these estimators satisfy, with exact sampling and arithmetic,

$$\begin{aligned} \mathbb{E}[Y_{c,a}] &= \mu_q(a \mid c), \\ \mathbb{E}[Y'_{j,c,a}] &= \partial_{\theta_j} \mu_q(a \mid c). \end{aligned} \tag{16}$$

The norm bounds $\|A_a\| = 1$ and $\|B_{ja}\| \le 2$ give second moments bounded by one and four [16, Sec. 3.4]. We evaluate ratios only at sampled points with nonzero reference amplitude. In particular,

$$\mathbb{E}[|W_{c,a}|^2] = \sum_{z : \psi_c(z) \neq 0} |\psi_c(z \oplus a)|^2 \le 1. \tag{17}$$

Averaging $S$ independent samples therefore gives mean-square errors at most $1/S$ and $4/S$ for a specified moment and phase derivative. By Chebyshev's inequality, $O(\epsilon^{-2})$ samples suffice for additive error $\epsilon$ at fixed confidence.

In practice, we draw $S$ independent reference samples $z_s \sim |\psi_c|^2$ and estimate the moment as

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s;\boldsymbol{\theta}). \tag{18}$$

We estimate the phase derivative by averaging the corresponding $Y'_{j,c,a}$ values over the same samples. For the standard IQP input $|0\rangle^{\otimes n}$, all reference amplitudes are equal, so $R_c(z,a) = 1$. The moment estimator then reduces to $Y = \cos[\Delta_a f_{\boldsymbol{\theta}}(z)]$, recovering the usual IQP cosine estimator [9, 21].

For a circuit with $r$ ancillary qubits, we apply the same estimator to the joint reference state, provided it supports the sampling-and-ratio interface. If we discard the ancillary measurement outcomes, we extend each visible-bit mask $a$ to $(a, 0^r)$, where $0^r$ denotes $r$ zeros. The zero entries ensure that the observable acts as the identity on the ancillary qubits.

## III. CONDITIONAL IQP TRAINING

Section II provides classical estimators of conditional Pauli-$Z$ moments and their phase gradients. We now combine these estimators with the moment formulation of Gaussian-kernel squared MMD to train one shared IQP unitary across conditions. Throughout this work, we use *classically trainable* to mean that we can efficiently estimate the training objective and its phase gradients to a prescribed additive accuracy on a classical computer.

### A. Condition-balanced MMD objective

We optimize the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions $p$ and $q$ on $\{0,1\}^n$ and a positive-definite kernel $k$, the population squared MMD is [10]

$$\begin{aligned}\mathrm{MMD}^2(p,q) = \mathbb{E}_{x,x'\sim p}[k(x,x')] - 2\mathbb{E}_{x\sim p,\, y\sim q}[k(x,y)] \\ + \mathbb{E}_{y,y'\sim q}[k(y,y')],\end{aligned} \tag{19}$$

where we draw the two samples in each expectation independently.

Throughout this work, we use the Gaussian kernel on binary strings,

$$k_\sigma(x,y) = \exp\left[-\frac{d_H(x,y)}{2\sigma^2}\right], \tag{20}$$

Here, $d_H(x,y)$ is the Hamming distance, which counts the positions at which $x$ and $y$ differ, and $\sigma > 0$ is the kernel bandwidth. For binary strings, $d_H(x,y) = \|x - y\|_2^2$, so this is the standard Gaussian kernel on the binary domain. For any finite positive $\sigma$, its population squared MMD vanishes exactly when $p = q$ [10].

Let $\mathcal{C}_{\mathrm{train}}$ denote the set of training conditions and $\tau$ the set of kernel bandwidths. We train the shared model by minimizing the condition-balanced objective

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}_{\mathrm{train}}|}\sum_{c\in\mathcal{C}_{\mathrm{train}}}\frac{1}{|\tau|}\sum_{\sigma\in\tau}\mathrm{MMD}^2_\sigma(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c))\,. \tag{21}$$

Here, $\mathrm{MMD}^2_\sigma$ denotes the squared MMD computed with the kernel $k_\sigma$. We weight all training conditions and bandwidths equally. At each optimizer update, we independently draw one condition uniformly from $\mathcal{C}_{\mathrm{train}}$ and one bandwidth uniformly from $\tau$, giving a stochastic estimate of this average.

### B. Character expansion

We use the Gaussian-kernel expansion into binary characters employed in earlier Born-machine and classical IQP training work [11, 12]. We apply this expansion separately at each condition to express the shared training objective through conditional Pauli-$Z$ moments. We sample a visible-bit mask $a \in \{0,1\}^n$ by drawing its coordinates independently according to

$$a_i \sim \mathrm{Bernoulli}(p_\sigma), \qquad p_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}, \tag{22}$$

The corresponding binary character is

$$\chi_a(x) = (-1)^{a\cdot x}, \tag{23}$$

which records the parity of the bits selected by $a$. For each bit where $x$ and $y$ agree, the expected character-product factor is 1. Where they differ, it is $1 - 2p_\sigma = e^{-1/(2\sigma^2)}$. Since the mask coordinates are independent, multiplying these factors recovers the Gaussian kernel:

$$k_\sigma(x,y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]\,. \tag{24}$$

For a condition $c$, define the target and model character moments

$$\begin{aligned}\mu_p(a\mid c) &= \mathbb{E}_{x\sim p(\cdot|c)}[\chi_a(x)], \\ \mu_q(a\mid c) &= \mathbb{E}_{x\sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)].\end{aligned} \tag{25}$$

Substituting Eq. (24) into Eq. (19) gives

$$\mathrm{MMD}^2_\sigma(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c)) = \mathbb{E}_a\Big[(\mu_p(a\mid c) - \mu_q(a\mid c))^2\Big]\,. \tag{26}$$

Thus, the squared MMD averages the squared differences between target and model character moments over the mask distribution.

For a Born machine, the visible Pauli operator

$$Z_a = \bigotimes_{i=1}^n Z_i^{a_i} \tag{27}$$

has eigenvalue $\chi_a(x)$ on the computational-basis state $|x\rangle$. Its model expectation value is therefore

$$\mu_q(a\mid c) = \langle Z_a\rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}. \tag{28}$$

We can thus express the conditional squared MMD entirely in terms of visible Pauli-$Z$ moments. We estimate the model moments classically using the reference-state estimator in Sec. II A.

The bandwidth $\sigma$ determines which correlation orders the objective emphasizes. The mask's Hamming weight $K = |a|$ counts the bits it selects, so it gives the correlation order of $Z_a$. Because we sample the mask coordinates independently,

$$\begin{aligned}K &\sim \mathrm{Binomial}(n, p_\sigma), \\ \mathbb{E}[K] &= np_\sigma, \\ \mathrm{Var}(K) &= np_\sigma(1-p_\sigma).\end{aligned} \tag{29}$$

Each bandwidth therefore induces a distribution over correlation orders. Large bandwidths favor lower-order

correlations, whereas decreasing $\sigma$ shifts weight toward higher-order correlations, with $\mathbb{E}[K] \to n/2$ as $\sigma \to 0$.

The bandwidth can also influence optimization behavior. Under suitable circuit and initialization assumptions, objectives dominated by global terms can exhibit barren plateaus, in which gradients vanish exponentially with system size [22–24]. These optimization difficulties are distinct from the efficient classical estimation of objectives and gradients that we call classical trainability. Emphasizing lower-order correlations can mitigate vanishing gradients in some settings, but it reduces sensitivity to higher-order differences between distributions [11]. For every finite positive bandwidth, the Gaussian kernel retains all correlation orders with positive weight. Increasing the bandwidth therefore changes their relative emphasis rather than removing them. In contrast, restricting an objective to low-order moments can make distinct distributions indistinguishable. Proposed approaches to improving optimization include tailored generator sets, restricted initializations and adversarial kernel adaptation [25–27]. Determining how these approaches extend to our conditional framework remains beyond the scope of this work.

### C. Unbiased finite-sample MMD and gradients

For a fixed condition $c$ and visible mask $a$, we draw $B \geq 2$ independent target samples and compute

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot \mid c). \tag{30}$$

Independently of these target samples, we draw $S \geq 2$ independent reference samples and evaluate the reference-state estimator from Eq. (12),

$$Y_s = Y_{c,a}(z_s; \boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2. \tag{31}$$

We denote the corresponding sample means by

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{32}$$

We follow the independent-pair construction used for unbiased IQP MMD estimation [12, Appendix A.2.1], applying it to the conditional reference estimators and target moments.

The direct estimate $(\overline{Y}-\overline{T})^2$ overestimates the squared moment difference on average because it includes the sampling variances of both sample means. We remove this bias by excluding products that use the same draw twice:

$$\begin{aligned}\widehat{\Delta^2_c}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_s Y_t - 2\overline{Y}\,\overline{T} \\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_b T_{b'}.\end{aligned} \tag{33}$$

Independence within each pair gives

$$\mathbb{E}\left[\widehat{\Delta^2_c}(a)\right] = \left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)^2. \tag{34}$$

Thus, Eq. (33) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (26). An unbiased finite-sample estimate of $\mathrm{MMD}^2$ may be negative.

Once we have computed the sample values, we evaluate the estimator in time linear in the numbers of target and reference samples. Appendix A 2 gives the pair-sum identities used for this calculation.

The same construction gives an unbiased estimator of the phase gradients. For a phase parameter $\theta_j$, let

$$Y'_s = \partial_{\theta_j} Y_{c,a}(z_s; \boldsymbol{\theta}). \tag{35}$$

We evaluate $Y'_s$ using Eq. (14). Differentiating Eq. (33) gives

$$\partial_{\theta_j}\widehat{\Delta^2_c}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_s Y'_t - 2\overline{T}\,\overline{Y'}, \tag{36}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_{s} Y'_s. \tag{37}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta^2_c}(a)\right] = 2\left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)\partial_{\theta_j}\mu_q(a \mid c), \tag{38}$$

which is the gradient of the corresponding squared moment difference.

Since $|T_b| = 1$, independence of distinct draws and the triangle inequality in $L^2$ give

$$\mathbb{E}\left[\left(\widehat{\Delta^2_c}(a)\right)^2\right] \leq 16,$$
$$\mathbb{E}\left[\left(\partial_{\theta_j}\widehat{\Delta^2_c}(a)\right)^2\right] \leq 64.$$

Averaging over conditions, bandwidths and sampled masks preserves these uniform bounds. Therefore, averaging $N$ independent repetitions of the complete estimator gives mean-square error at most $16/N$ for the objective and $64/N$ for a specified phase-gradient component. By Chebyshev's inequality, $O(\epsilon^{-2})$ repetitions suffice for additive error $\epsilon$ at fixed confidence.

At a fixed parameter vector, the expectations above average over independent draws from the target distribution and the reference sampling law. During training, we sample target minibatches with replacement from fixed training pools; the stochastic gradient is therefore unbiased for the empirical objective associated with those pools.

We exclude the diagonal terms $s = t$ because $Y_s$ and $Y'_s$ can be correlated when we compute them from the same reference draw. The fixed encoder makes the reference sampling law independent of $\boldsymbol{\theta}$. We therefore hold the sampled reference strings fixed during automatic differentiation to evaluate the phase gradients without differentiating through the sampler.

### D. Stochastic mask estimator

Equation (26) expresses the squared MMD as an expectation over the mask distribution in Eq. (22). Since there are $2^n$ possible masks, we estimate this expectation using a finite sample of masks.

The identity mask $a = 0$ satisfies

$$\mu_p(0 \mid c) = \mu_q(0 \mid c) = 1 \tag{39}$$

for every condition and therefore contributes zero to the squared MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{40}$$

Under the original mask distribution, nonidentity masks have total probability

$$Q_\sigma^+ = 1 - (1 - p_\sigma)^n. \tag{41}$$

We multiply the conditional expectation by $Q_\sigma^+$ to preserve the original Gaussian kernel objective.

We draw $M$ nonidentity masks $a_1, \ldots, a_M$ independently from $P_\sigma(a \mid a \neq 0)$ and estimate the squared MMD at condition $c$ as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \frac{1}{M} \sum_{m=1}^{M} \widehat{\Delta}_c^2(a_m), \tag{42}$$

Here, $\widehat{\Delta}_c^2$ is the unbiased finite-sample estimator in Eq. (33). For each sampled condition and bandwidth, we use the same target and reference batches for all masks, drawing these batches independently of the masks. Sharing batches correlates the estimates across masks, but linearity of expectation preserves unbiasedness. Averaging over the mask, target and reference draws therefore recovers the full kernel squared MMD.

Repeated masks use the same target and reference batches, so they yield identical estimates and phase gradients. We therefore evaluate each distinct mask once and weight its contribution by its frequency among the $M$ draws. This preserves the sampled objective and its gradient, as detailed in Appendix A 2.

## IV. NUMERICAL EXPERIMENTS

We use numerical experiments to assess the capabilities of the conditional framework. Translation of Gaussian-shaped profiles tests the basic conditioning mechanism, while two-coordinate conditioning tests joint control of profile center and width at combinations absent from training. The Curie–Weiss model tests whether one shared unitary can reproduce temperature-dependent changes in distribution shape. These benchmarks use sixteen qubits. Finally, training runs from 64 to 2048 qubits assess computational feasibility. For the sixteen-qubit benchmarks, we use the exact full-distribution total variation distance as the primary accuracy metric,

$$\mathrm{TVD}(p, q) = \frac{1}{2} \sum_{x \in \{0,1\}^n} |p(x) - q(x)|. \tag{43}$$

We also report the exact TVD between Hamming-weight profiles as a diagnostic,

$$\mathrm{TVD}_K(p, q) = \frac{1}{2} \sum_{k=0}^{n} |\rho_p(k) - \rho_q(k)|, \tag{44}$$

where $\rho_p(k)$ and $\rho_q(k)$ give the probabilities of observing Hamming weight $k$ under $p$ and $q$, respectively. Grouping bitstrings by Hamming weight gives $\mathrm{TVD}_K(p, q) \leq \mathrm{TVD}(p, q)$. The Gaussian-shaped and Curie–Weiss targets assign equal probability to all bitstrings with the same Hamming weight, but the learned models need not preserve this property. Profile TVD therefore does not resolve errors within Hamming-weight shells. We compute both metrics using the same exact circuit probabilities.

We train the models by minimizing the squared-MMD objective. We use hyperparameter optimization for the two-coordinate, Curie–Weiss and scaling benchmarks. Appendices A and B specify the training settings, hyperparameter searches and evaluation protocols. Code and data are available in the accompanying repository [28].

### A. Translation of Gaussian-shaped profiles

Our first experiment tests whether one shared unitary converts condition-dependent input phases into translated output profiles. For $n = 16$ bits, let $K = |x|$ denote Hamming weight. We define a Gaussian-shaped family by tilting a binomial profile,

$$\rho(k \mid c) \propto \binom{16}{k} \exp\left[\frac{3}{4}(2c - 1)(k - 8)\right], \tag{45}$$

for $k \in \{0, \ldots, 16\}$ and $c \in [0, 1]$. The target bitstring law is $p(x \mid c) = \rho(|x| \mid c)/\binom{16}{|x|}$. At $c = 0.5$, it is exactly uniform and $K$ is binomial. The linear tilt gives a binomial profile with condition-dependent success probability. It shifts the mean and also changes the standard deviation. These profiles are Gaussian-shaped, rather than exact discrete Gaussians of fixed width.

The product-state encoder stores $c$ only in phases,

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c - 1/2) R_Y(\pi/2) \, |0\rangle_i \, . \tag{46}$$

Measuring this input always gives a uniform bitstring distribution. The learned IQP unitary must therefore convert phases into the nonuniform target probabilities away from the center. At $c = 0.5$, however, the input

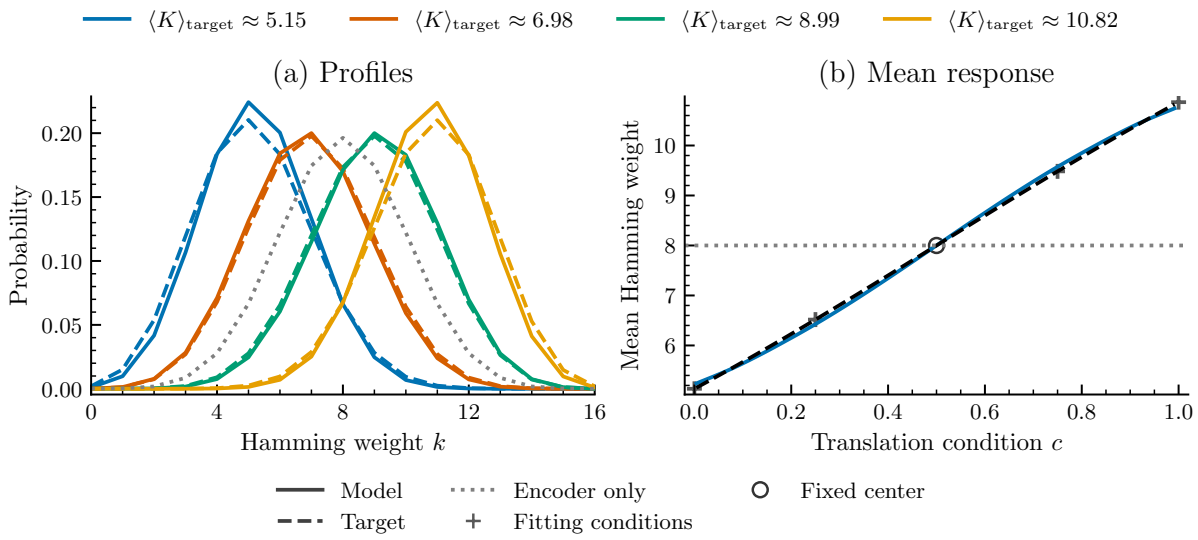



FIG. 2. Conditional translation of Gaussian-shaped Hamming-weight profiles at sixteen qubits. (a) Solid colored curves show exact model profiles averaged across five fits at four held-out conditions spread throughout $[0, 1]$; dashed curves show their targets. The upper labels identify the target mean Hamming weight for each color. (b) The solid blue curve and shading show the median and interquartile range of the model's mean output Hamming weight at 80 held-out conditions. The dashed black curve shows the exact target response. Dotted gray curves show the uniform encoder's binomial profile and constant mean; gray markers denote fitting conditions. The open circle at $c = 0.5$ marks exact agreement imposed by construction. Table I reports full-distribution and profile errors.

is $|+\rangle^{\otimes 16}$, an eigenstate of every $X$-type IQP generator. Every parameter vector preserves its uniform output law, so agreement at this condition is architectural and does not provide evidence of learning.

We use a $4 \times 4$ lattice with sixteen one-body and twenty-four nearest-neighbor two-body generators, each with an independent learned parameter. We fit at $c \in \{0, 0.25, 0.5, 0.75, 1\}$ and evaluate at 80 held-out conditions spaced throughout $[0, 1]$. Appendix A 3 specifies sampled training and exact development selection.

The median across five fits of mean full-distribution TVD over these reporting conditions is 0.0668, compared with 0.2848 for the encoder alone. Profile TVD is 0.0212. Figure 2 shows how the shared unitary shifts the profiles and follows the target mean between fitting conditions. The larger full-distribution error reveals discrepancies that Hamming-weight aggregation conceals. These results validate the conditioning mechanism away from the fixed central baseline.

TABLE I. Exact full-distribution and profile TVDs for Gaussian-shaped translation. Entries report the median across five independent fits of the mean error over eighty reporting conditions; brackets give the 25th–75th percentiles across fits. Lower is better.

| Model | Full TVD | Profile TVD |
|---|---|---|
| Conditional IQP | 0.0668 [0.0624, 0.0692] | 0.0212 [0.0171, 0.0219] |
| Encoder only | 0.2848 | 0.2848 |

### B. Two-coordinate conditioning and generalization

We next test joint control of profile center and width, including coordinate combinations absent from fitting. We extend Eq. (45) to

$$p(x \mid u, v) = \frac{\rho_{u,v}(|x|)}{\binom{16}{|x|}}, \tag{47}$$

$$\rho_{u,v}(k) \propto \binom{16}{k} \exp\left[-\frac{(k - 8 - 3u)^2}{8(1 + 0.2v)^2} + \frac{(k-8)^2}{8}\right], \tag{48}$$

where $u, v \in [-1, 1]$ and we normalize over $k \in \{0, \ldots, 16\}$. The previous family is the slice $u = 2c - 1$, $v = 0$. The linear and quadratic tilts control center and width, respectively, but their finite-support moments also depend on one another. We therefore compare generated responses with actual target means and standard deviations. At $(u, v) = (0, 0)$, the target is exactly uniform.

We encode the coordinates through collective phases on a uniform input. With $J_z = \frac{1}{2}\sum_j Z_j$, the encoder prepares

$$|\phi(u, v)\rangle = e^{-i(u/2)J_z} e^{-i(v/40)J_z^2} |+\rangle^{\otimes 16}. \tag{49}$$

The linear phase gives a constant phase increment between neighboring Hamming-weight sectors. The quadratic phase makes this increment depend on Hamming weight and introduces correlations for nonzero $v$. These phases provide condition-dependent features,

while every input has the same uniform computational-basis probabilities. Consequently, the encoder does not prepare the requested noncentral distributions. One shared IQP transformation must convert the features into changes in the output law.

The learned layer uses the same forty independently parameterized lattice generators as translation. Its parameters need not preserve permutation symmetry, so full-distribution TVD can exceed profile TVD. Only the encoder and reference state retain seventeen distinct weight-dependent amplitudes. This structure permits polynomial-cost reference sampling and amplitude-ratio evaluation, as detailed in Appendix B. At the central condition, every learned unitary preserves the uniform law. We design the target family to share this fixed baseline and assess learned responses away from it.

We compare two fitting geometries. Axis training varies $u$ at $v = 0$ and $v$ at $u = 0$, excluding mixed pairs from fitting and development selection. Reporting then tests new combinations within and beyond the diamond-shaped fitting hull, while remaining within each observed coordinate range. Grid training fits $\{-1, 0, 1\}^2$ and tests interpolation within its rectangular hull. Both designs use nine fitting conditions and five independent final fits. We select optimizer settings and checkpoints using exact development TVD, with protocol details in Appendix B.

Axis training gives mean reporting full-distribution TVD $0.0468 \pm 0.0143$ over sixteen unseen mixed pairs. Four lie within its fitting hull and give $0.0185 \pm 0.0060$; twelve lie outside and give $0.0563 \pm 0.0171$. Grid training gives $0.0397 \pm 0.0093$ over thirty-two withheld pairs. We average errors over conditions within each fit, then report the mean and sample standard deviation across fits. Mean-response sweeps follow the target displacement across all tested widths. Every final fit increases width along each tested slice, although the amount of broadening varies across fits. Appendix B shows the mean and width responses and distinguishes full-distribution errors from profile errors.

These experiments test a shared transformation across a task-informed family. They demonstrate responses at combinations absent from fitting and development selection, but do not establish that the lattice is necessary or that the selected encoder generalizes to unrelated families. The condition sets and phase features were explored during earlier design, and exact development targets contribute to selection. We therefore interpret the results as conditional capability within the studied domain.

### C. Curie–Weiss spin model

We use the zero-field Curie–Weiss model [29] to test conditional generation across a change in distribution shape. For $n = 16$ spins $s_i \in \{-1, +1\}$ with unit interaction strength, the target distribution is

$$p(x \mid \beta) = \frac{1}{\mathcal{Z}_\beta} \exp\left[\frac{\beta}{2n}(2|x| - n)^2\right], \tag{50}$$

where $\beta$ is inverse temperature and $\mathcal{Z}_\beta$ normalizes the distribution. The target assigns equal probability to bit-strings with the same Hamming weight and is invariant under global spin flip. As $\beta$ increases, its finite-size Hamming-weight profile changes from a central peak to two symmetric modes.

We encode temperature through correlated input phases chosen to preserve the target's global spin-flip symmetry:

$$|\phi_{\mathrm{P}}(\beta; s)\rangle = \exp\left[-\frac{is\beta}{2}\sum_{i=1}^{15} Z_i Z_{16}\right] |+\rangle^{\otimes 16}, \tag{51}$$

where $s$ is an encoder scale selected by hyperparameter optimization. These pairwise rotations introduce temperature-dependent phase correlations, but leave the computational-basis probabilities uniform at every temperature. Thus the encoder supplies conditional information without producing the target family itself. The learned IQP unitary must convert this information into the different target distributions through one shared transformation across temperatures.

A preparation circuit applies $R_X(s\beta)$ to each of the first fifteen qubits initially in $|0\rangle$, copies their parity to qubit sixteen using CNOTs, and applies a Hadamard to every qubit. This construction uses no ancillas and has a classically tractable reference state $|\psi_{\mathrm{P}}\rangle = H^{\otimes 16} |\phi_{\mathrm{P}}\rangle$.

Let $u \in \{0, 1\}^{15}$ and let $b$ denote the final bit. The reference amplitudes are

$$\psi_{\mathrm{P}}(u, b) = \mathbf{1}_{b=\oplus_i u_i} \cos(s\beta/2)^{15-|u|} [-i \sin(s\beta/2)]^{|u|}. \tag{52}$$

We sample the first fifteen bits independently with probability $\sin^2(s\beta/2)$ and set the final bit to their XOR. Reference amplitude ratios follow from the displayed product on its support and vanish when the mask has odd parity. These operations meet the sampling-and-ratio interface in $O(n)$ arithmetic operations for the corresponding $n$-qubit construction.

The reference has even total parity, so $Z^{\otimes 16} |\psi_{\mathrm{P}}\rangle = |\psi_{\mathrm{P}}\rangle$ and $X^{\otimes 16} |\phi_{\mathrm{P}}\rangle = |\phi_{\mathrm{P}}\rangle$. Every generator in the learned IQP unitary commutes with $X^{\otimes 16}$, giving

$$q_{\boldsymbol{\theta}}(x \mid \beta) = q_{\boldsymbol{\theta}}(\bar{x} \mid \beta) \tag{53}$$

for every parameter vector, where $\bar{x}$ flips all bits. This guarantees zero signed mean magnetization and a symmetric Hamming-weight profile. It does not guarantee permutation invariance, uniformity within shells or agreement with the target distribution.

We use all one- and two-body IQP generators, giving 136 independent learned phases per circuit. The shared model fits $\beta \in \{0.1, 0.7, 1.3, 1.95\}$ and reports at the common markers $\beta \in \{0.4, 1, 2\}$ and on a separate dense grid. Each independent baseline family contains three circuits fitted directly at the markers. The independent parity-input family uses Eq. (51), with one jointly selected scale and optimizer configuration across its circuits. The independent unconditional family uses $|0\rangle^{\otimes 16}$ and does not

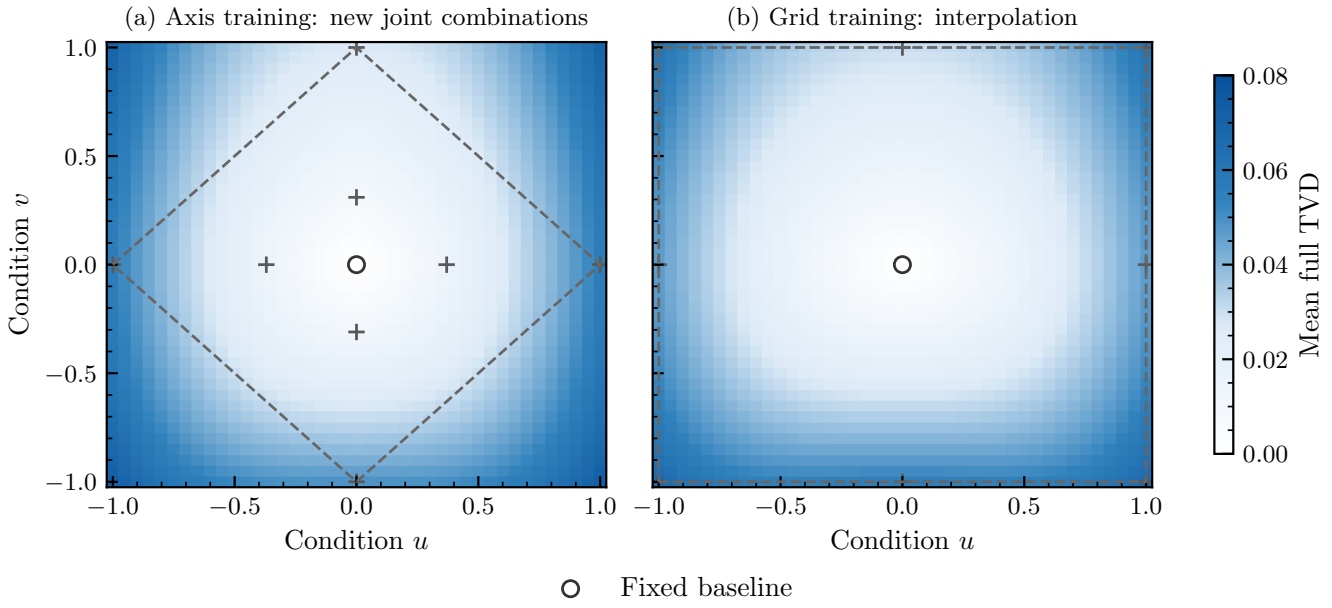



FIG. 3. Two-coordinate conditional generation with forty independent lattice parameters. Blue shading shows the pointwise mean full-distribution TVD across five fits on a $41 \times 41$ diagnostic grid. Dark gray crosses mark fitting conditions and gray dashed lines delimit their convex hulls. (a) Axis training tests new combinations within and beyond the diamond-shaped hull; all displayed conditions remain within the individual coordinate ranges seen during fitting. (b) Grid training tests interpolation within the rectangular hull. Both panels share one color scale. The open circle at $(u, v) = (0, 0)$ marks the uniform target and model law fixed by construction. Dense sweeps include fitting and development conditions; reporting averages use separate sets.

TABLE II. Two-coordinate reporting errors. Values are means and sample standard deviations of per-fit mean errors across five independent fits. Full TVD is the primary metric and selects checkpoints on separate development conditions. The two fitting geometries have different reporting sets, so columns do not form a paired accuracy comparison. Lower is better.

| | Axis fitting, sixteen pairs | Grid fitting, thirty-two pairs |
|---|---|---|
| Encoder only, full and profile TVD | 0.3867 | 0.3586 |
| Learned model, full TVD | $0.0468 \pm 0.0143$ | $0.0397 \pm 0.0093$ |
| Learned model, profile TVD | $0.0173 \pm 0.0066$ | $0.0156 \pm 0.0036$ |

impose global spin-flip symmetry. Appendix A 4 b compares the shared parity model with a shared product-state encoder using the same learned IQP ansatz. This comparison assesses the encoder construction as a whole, since the replacement changes both input correlations and the enforced spin-flip symmetry.

We train with sampled squared MMD and perform hyperparameter optimization separately for each model family. For checkpoint and hyperparameter selection, we use exact target probabilities, with full-distribution TVD as the primary criterion and physical-observable agreement as a secondary criterion. Appendix A 4 a specifies the optimization and selection protocols. We report means and sample standard deviations over ten independent final repeats, retaining all outcomes.

Figure 4 shows that the shared parity model reproduces the change from a central magnetization profile to a strongly polarized bimodal profile, although shape discrepancies remain near $\beta = 1$. At the cold marker $\beta = 2$, its mean full-distribution TVD is $0.0832 \pm 0.0103$, compared with $0.1525 \pm 0.0021$ for independent parity, a 45.4% reduction. This advantage depends on temperature: independent parity performs better at $\beta = 0.4$, while shared parity has modestly lower full-distribution TVD at $\beta = 1$ despite larger profile error. Table III gives the complete marker comparison.

Across the separate 101-point reporting grid in $[0, 2]$, shared parity achieves inverse-temperature-averaged TVD $0.1033 \pm 0.0036$ and per-repeat maximum TVD $0.1406 \pm 0.0037$. Its mean error peaks near $\beta = 0.98$, placing the largest discrepancies near the crossover. The reporting markers test interpolation and, at $\beta = 2$, slight extrapolation beyond the largest fitting temperature 1.95. These conditions are excluded from shared fitting and selection, although earlier studies examined this temperature range. At $\beta = 0$, the encoder pre-

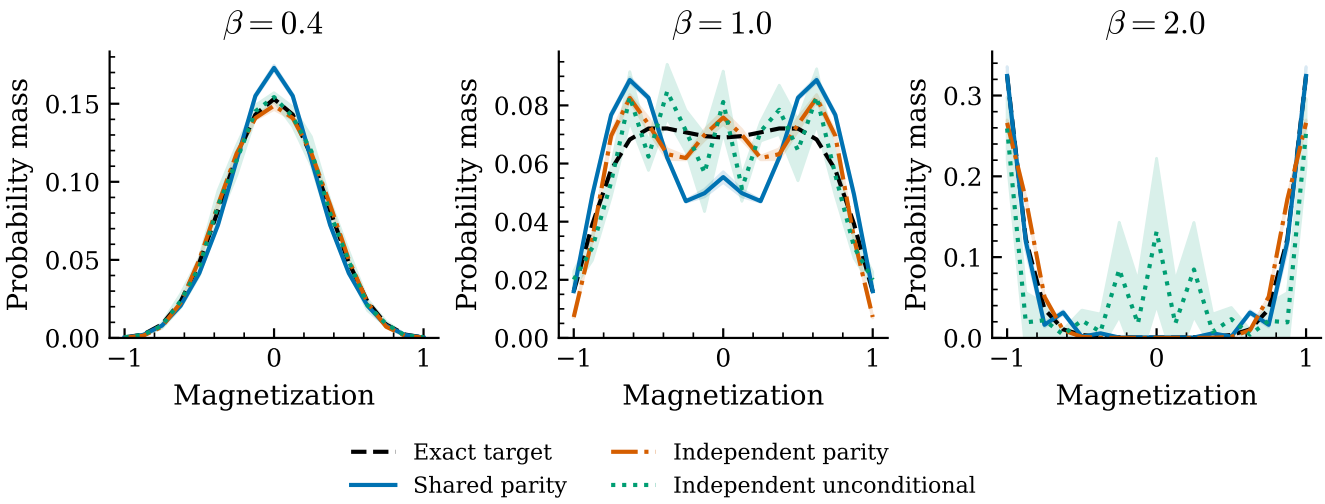



FIG. 4. Curie–Weiss magnetization profiles at sixteen qubits. The horizontal axis shows magnetization per spin, $m = (2k - 16)/16$, where $k$ is the Hamming weight of a bitstring. Model curves show mean exact Hamming-weight probability masses over ten fresh final repeats, and shaded bands show sample standard deviations. The dashed black curve is the exact target. The shared parity model excludes the displayed markers from fitting and selection. Independent circuits fit directly at each marker. Full-distribution TVD in Table III also measures errors within Hamming-weight shells.

TABLE III. Exact full-distribution TVD at sixteen qubits. Values are means ± sample standard deviations over ten fresh final repeats, computed within each repeat before aggregation. Lower is better. Shared parity excludes markers from fitting and selection; independent models fit directly at each marker.

| Model | $\beta = 0.4$ | $\beta = 1$ | $\beta = 2$ |
|---|---|---|---|
| Shared parity | $0.1185 \pm 0.0107$ | $0.1396 \pm 0.0048$ | $0.0832 \pm 0.0103$ |
| Independent parity | $0.0692 \pm 0.0059$ | $0.1477 \pm 0.0081$ | $0.1525 \pm 0.0021$ |
| Independent unconditional | $0.3787 \pm 0.0034$ | $0.3400 \pm 0.0074$ | $0.4101 \pm 0.1629$ |

pares $|+\rangle^{\otimes 16}$, which the learned unitary changes only by a global phase. The uniform output at this endpoint therefore follows from the architecture independently of training.

We also assess whether the learned distributions reproduce changes in physical observables. With $m = n^{-1} \sum_i s_i$, we use mean absolute magnetization $\langle |m| \rangle$, since the parity model enforces zero signed mean magnetization. For $H = -nm^2/2$, we define the per-spin energy-fluctuation statistic

$$c_{\mathrm{fluc}}(\beta) = \frac{\beta^2}{n} \big( \langle H^2 \rangle - \langle H \rangle^2 \big). \tag{54}$$

For the Gibbs target, this equals the equilibrium heat capacity per spin. A learned distribution need not satisfy the corresponding temperature-derivative identity. Appendix A 4 d gives the observable calculations.

Figure 5 shows that shared parity captures the broad magnetization change and fluctuation peak. However, observable agreement does not follow the full-distribution TVD ranking. At $\beta = 1$, independent parity reproduces mean absolute magnetization more accurately, whereas shared parity better matches energy fluctuations. At $\beta = 2$, shared parity has lower TVD but larger fluctuation error. The independent unconditional model also shows substantial variation across repeats at this cold marker. These results demonstrate conditional distribution learning while identifying limitations in the recovered physical observables. Table X reports the complete observable-error comparison.

### D. Scaling

We next assess the computational feasibility of classical conditional training beyond sixteen qubits. We train Curie–Weiss models at $n \in \{64, 128, 256, 512, 1024, 2048\}$, without ancillas, using the parity encoder introduced above. This encoder preserves global spin-flip symmetry at every system size. The learned unitary contains independent one-body phases and two-body phases between horizontal and vertical nearest neighbors on an open rectangular lattice. An $L_x \times L_y$ lattice therefore contains $3n - L_x - L_y$ trainable parameters, giving linear parameter growth.

We perform hyperparameter optimization at 64 and 256 qubits, then use the same selected configuration at all six sizes. At each size, we train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$

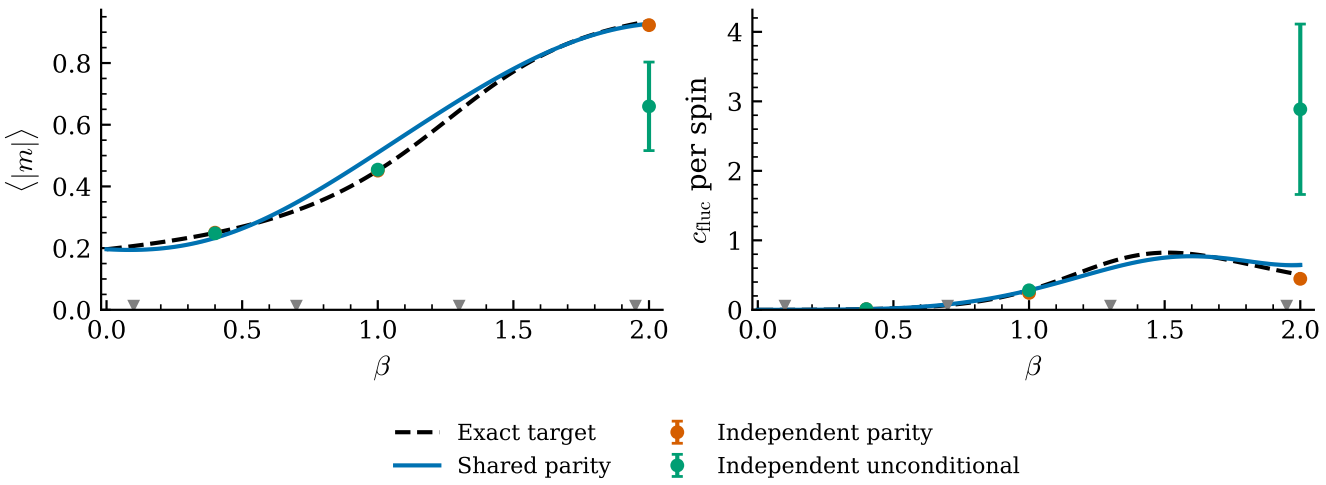



FIG. 5. Mean absolute magnetization and energy fluctuations for sixteen-qubit Curie–Weiss models. The dashed black curves show the exact target. Shared parity curves and shaded bands show means and sample standard deviations over ten fresh final repeats on the fixed reporting grid. Independent parity and unconditional models appear as marker means with sample-standard-deviation error bars at their directly fitted temperatures $\beta = 0.4, 1, 2$. Grey triangles mark the four shared fitting anchors. Each observable is computed within each repeat before aggregation. Equation (54) defines the fluctuation statistic for learned distributions; it does not verify a thermal derivative.

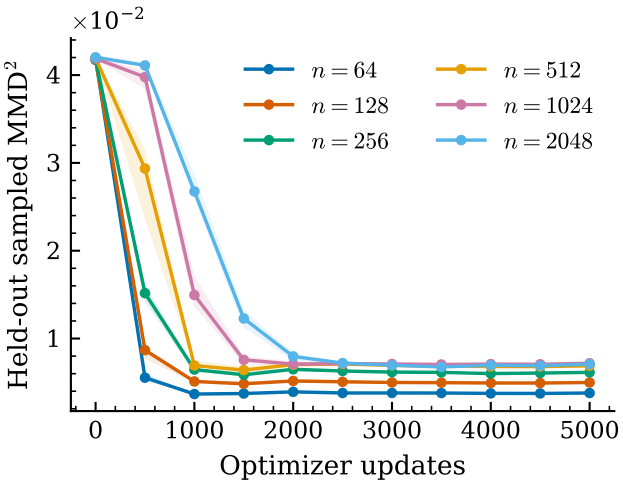



FIG. 6. Classical training of conditional IQP Born machines with the parity encoder from 64 to 2048 qubits. At each plotted update, we average sampled MMD$^2$ over the held-out inverse temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. For each run, we show the trajectory of its selected initialization. Curves and shaded bands show the median and interquartile range across the three training runs. We fix the expected mask weights at $w \in \{2, 4\}$, so the corresponding kernel bandwidths vary with system size.

and track sampled MMD$^2$ at held-out temperatures. Appendix A 5 specifies the architectures, optimization and evaluation protocols.

Figure 6 shows a decrease in held-out sampled MMD$^2$ at every size, followed by a plateau. After 5000 updates, the median decreases from 0.0419 to 0.00379 at 64 qubits and from 0.0420 to 0.00708 at 2048 qubits. These results demonstrate classical conditional training with a symmetry-preserving encoder up to 2048 qubits, with progress on conditions excluded from fitting and selection. However, a low sampled MMD$^2$ does not by itself establish high full-distribution fidelity. The kernel retains all correlation orders, but the fixed expected mask weights $w \in \{2, 4\}$ emphasize low-order Pauli-$Z$ moments. Discrepancies in higher-order correlations may therefore remain weakly penalized or poorly resolved by finite sampling. Moreover, the size-dependent kernel bandwidths prevent direct fidelity comparisons across system sizes.

## V. INPUT-STATE INDUCED SYMMETRIES IN IQP BORN MACHINES

Input-state encoding can also enforce exact output symmetries without restricting the learned phases. For a nonzero bit mask $s \in \{0, 1\}^n$, the bit-flip operator $X_s = \bigotimes_{i=1}^{n} X_i^{s_i}$ commutes with the IQP unitary. If each encoded input is an eigenstate of $X_s$, the output state retains the same eigenvalue. Its computational-basis amplitudes at $x$ and $x \oplus s$ therefore differ at most by a sign, giving equal probabilities for every condition and parameter vector. This symmetry principle underlies the input-state construction of Recio-Armengol *et al.* [12, Sec. 6]. Building on it, we construct condition-dependent encoders that enforce several prescribed bit-

flip symmetries while retaining efficient reference sampling and amplitude-ratio evaluation. Such constraints provide inductive biases in variational quantum models [30].

Our construction maps independent latent bits to reference bitstrings that satisfy specified parity constraints. The Hadamard layer then converts these constraints into bit-flip symmetries. Specifically, let $h_1, \ldots, h_r \in \mathbb{F}_2^n$ be linearly independent symmetry masks. Choose a matrix $C \in \mathbb{F}_2^{n\times k}$, with $k = n - r$, whose columns form a basis of their common orthogonal subspace. Thus, $h_j^\top C = 0$ for every prescribed mask, and $C$ maps the $k$ latent bits to an allowed $n$-bit reference configuration. All matrix operations below are over $\mathbb{F}_2$.

For each condition $c$, choose efficiently evaluable factors satisfying

$$|f_\ell(0;c)|^2 + |f_\ell(1;c)|^2 = 1. \tag{55}$$

Define

$$\begin{aligned} |\psi_c\rangle &= \sum_{u\in\mathbb{F}_2^k} \left[\prod_{\ell=1}^{k} f_\ell(u_\ell;c)\right] |Cu\rangle\,, \\ |\phi(c)\rangle &= H^{\otimes n}\,|\psi_c\rangle\,. \end{aligned} \tag{56}$$

The matrix $C$ fixes the symmetry constraints, while the factors supply the condition dependence.

Reference sampling draws the latent bits independently with probabilities $|f_\ell(1;c)|^2$ and returns $Cu$. For a mask $a = Cd$, injectivity of $C$ gives a unique displacement $d$, and

$$R_c(Cu, a) = \prod_{\ell=1}^{k} \frac{f_\ell(u_\ell \oplus d_\ell; c)}{f_\ell(u_\ell; c)}. \tag{57}$$

If $a \notin \operatorname{im} C$, the ratio is zero. We evaluate denominators only at sampled configurations with nonzero amplitude.

Every $h \in \ker C^\top$ satisfies $Z_h\,|\psi_c\rangle = |\psi_c\rangle$, and hence $X_h\,|\phi(c)\rangle = |\phi(c)\rangle$. The learned IQP unitary preserves these eigenstates, giving

$$q_{\boldsymbol{\theta}}(x \mid c) = q_{\boldsymbol{\theta}}(x \oplus h \mid c) \qquad \text{for all } h \in \ker C^\top. \tag{58}$$

This relation holds for every condition and parameter vector. The construction therefore guarantees a subgroup of $2^r$ bit-flip symmetries. Particular factor functions or parameter values can produce additional symmetries.

The Curie–Weiss encoder in Eq. (51) uses $Cu = (u, \bigoplus_\ell u_\ell)$, which guarantees global spin-flip symmetry. The switching encoder in Section V A uses $C = B^\top$, where $B$ is the fundamental-cycle matrix defined in Appendix D, and guarantees the six independent vertex switches. Both constructions preserve the sampling-and-ratio interface. Their symmetry guarantees hold independently of parameter tying and training accuracy. Appendix C establishes normalization, access costs and state preparation.

### A. Signed-graph encoding with exact switching symmetry

The Curie–Weiss encoder enforces one global spin-flip symmetry. To illustrate how the binary-linear construction imposes several independent symmetries, we now consider signed graphs. Different edge-sign configurations can share the same triangle interactions, providing a concrete setting for symmetry-preserving conditional inputs.

Each edge carries a positive or negative sign. A vertex switch reverses the signs of all edges incident on that vertex. However, it preserves every triangle product because it reverses either zero or two signs within each triangle. Distributions governed by these products therefore remain invariant under vertex switches. For the seven-vertex graph considered here, six independent switches generate 64 symmetry transformations. We construct a temperature-dependent encoder that guarantees all these output symmetries for every IQP parameter vector.

Sixteen qubits represent the edge signs of a seven-vertex graph. We connect every pair among vertices $0, \ldots, 4$, and connect each of vertices 5 and 6 to vertices $0, 1, 2$. Each edge bit $x_{ij}$ specifies a sign $s_{ij} = (-1)^{x_{ij}}$. A Gibbs family at inverse temperature $\beta$ has the form

$$p_\beta(x) = Z_\beta^{-1} e^{\beta E(x)}, \qquad E(x) = \sum_{\{i,j,k\}\in\mathcal{T}} s_{ij}s_{ik}s_{jk}, \tag{59}$$

where $\mathcal{T}$ contains the graph's sixteen triangles and $Z_\beta$ normalizes the distribution. We call a triangle balanced when the product of its three edge signs is positive. Increasing $\beta$ therefore favors graphs with more balanced triangles.

We describe combinations of vertex switches using signs $h_i \in \{-1, +1\}$, where $h_i = -1$ indicates a switch at vertex $i$. Their combined action replaces each edge sign $s_{ij}$ by $h_i s_{ij} h_j$. Switching all seven vertices reverses every edge twice and leaves the graph unchanged. We therefore fix $h_0 = 1$ to remove duplicate representations. The remaining six binary choices give $2^6 = 64$ distinct switching transformations, including the identity. We denote their corresponding edge-bit masks by $G$; applying a mask $g \in G$ flips the indicated bits, $x \mapsto x \oplus g$. Every transformation preserves this Gibbs distribution at every temperature.

We encode temperature using ten triangles that share vertex 0. The six root edges $0i$, with $i = 1, \ldots, 6$, form a spanning tree: they connect all seven vertices without forming a cycle. We call the remaining ten edges *chords* and denote their set by $\mathcal{C}$. Each chord $ij$, together with root edges $0i$ and $0j$, forms a triangle $(0, i, j)$.

We initialize all sixteen qubits in $|0\rangle$. For each chord $ij$, we apply $R_X(t_\beta)$ to its qubit, then use that qubit to control two CNOT gates targeting root edges $0i$ and $0j$. On an isolated triangle initially in $|000\rangle$, this motif creates a superposition of $|000\rangle$ and $|111\rangle$ (Figure 7). In either component, each vertex touches zero or two edges

with bit value one, giving even incidence parity.

In the full graph, the triangles share root-edge qubits. We repeat the rotation-and-CNOT pattern for all ten chords, combining the shared root-edge bits modulo two. This preserves even incidence parity at every vertex. Finally, we apply a single Hadamard layer to all sixteen qubits.

After the Hadamard layer, the encoded input takes the form

$$|\phi_\beta\rangle = \exp\left[-\frac{it_\beta}{2}\sum_{(i,j)\in\mathcal{C}} Z_{0i}Z_{0j}Z_{ij}\right]|+\rangle^{\otimes 16}, \tag{60}$$

$$t_\beta = \frac{\pi}{2}\tanh(1.5\beta). \tag{61}$$

We use this fixed temperature map as a concrete example; the symmetry guarantee does not depend on its form. Each Pauli-$Z$ product contributes a phase determined by the edge-sign product of its triangle. Because these operators are diagonal in the computational basis, they leave the uniform measurement probabilities of $|+\rangle^{\otimes 16}$ unchanged. Temperature therefore enters entirely through correlated phases. An IQP transformation can convert these phases into changes in the output distribution, while preserving the imposed symmetries.

We prepare the encoded input using ten rotations, twenty CNOTs and sixteen Hadamards, without ancillary qubits. The state before the Hadamard layer provides the reference for classical estimation. To sample its measurement distribution, we draw the ten chord bits independently, assigning each a value of one with probability $\sin^2(t_\beta/2)$. We then set each root-edge bit $0i$ to the sum, modulo two, of the chord bits incident on vertex $i$. The chord bits also determine each supported amplitude through a product of ten rotation coefficients, allowing us to evaluate the required amplitude ratios directly. Appendix D gives the explicit expressions.

The Hadamard layer converts the reference state's even incidence parity into invariance under vertex switches:

$$X^g|\phi_\beta\rangle = |\phi_\beta\rangle, \qquad g \in G, \tag{62}$$

where $X^g$ flips the edge bits specified by the switching mask $g$. Every IQP layer generated by Pauli-$X$ strings commutes with these switching operators and therefore preserves this invariance, regardless of its learned parameters.

The resulting model satisfies

$$q_{\boldsymbol{\theta}}(x \mid \beta) = q_{\boldsymbol{\theta}}(x \oplus g \mid \beta), \qquad g \in G, \tag{63}$$

for every condition and parameter vector, without averaging the output over switches. Thus, the encoder enforces six independent switching symmetries while leaving the IQP phases unrestricted. This example demonstrates the construction and its classical-access requirements; it does not assess learning accuracy for the Gibbs family.

## VI. DISCUSSION

Our framework extends classically trainable IQP Born machines to distribution families by combining reference-state estimators of Pauli-$Z$ moments and their phase gradients with a conditional MMD objective. One learned IQP unitary then acts across conditions, and the encoder need not have IQP form. Efficient reference sampling and amplitude-ratio evaluation allow encoder design to incorporate conditional structure and exact symmetries while preserving efficient classical estimation of the objective and its gradients.

The Gaussian-shaped benchmarks test how one shared lattice transformation converts conditional phases into shifts and changes in width. Their encoder probabilities remain uniform at every condition, while the targets are nonuniform away from the origin. The two-coordinate experiment recovers target mean responses and increasing width at new joint combinations, with larger errors beyond the axis-fitting hull and variation in the amount of broadening. However, the family and encoder are task-informed, and exact development targets select configurations and checkpoints. These results demonstrate conditional capability within the studied family, rather than a necessity for forty independent phases or robustness to arbitrary encoder design. Central agreement is architectural and does not contribute to the reporting averages.

The Curie–Weiss benchmark then tests a more pronounced change in distribution shape. A shared unitary learns the temperature-dependent transition from a central peak to two separated peaks, while the parity encoder guarantees global spin-flip symmetry. The comparison with product inputs favors the correlated encoder under the tested protocol, but changes both correlations and encoder phase generators, so it does not isolate the effect of symmetry. Independently fitted models remain more accurate at some temperatures and for some physical observables, while requiring separate parameter sets and more total optimizer updates.

The signed-graph construction extends symmetry control from one global spin-flip symmetry to six independent vertex-switching symmetries. A fundamental-cycle reference supplies the required binary parity constraints, while independent chord sampling and product amplitudes retain efficient classical access. These guarantees hold for every condition and IQP parameter vector. This worked example establishes the encoder construction rather than empirical accuracy for the associated Gibbs family.

The methodological limitations concern distributional evaluation and the scope of the scaling evidence. Efficient, unbiased MMD estimation does not ensure accurate recovery of every relevant correlation [31]. Although the Gaussian kernel includes all correlation orders, its bandwidth weights them unequally, and finite sampling and optimization may fail to resolve weakly weighted discrepancies. Agreement in Hamming-weight profiles can therefore coexist with errors in individual bitstring

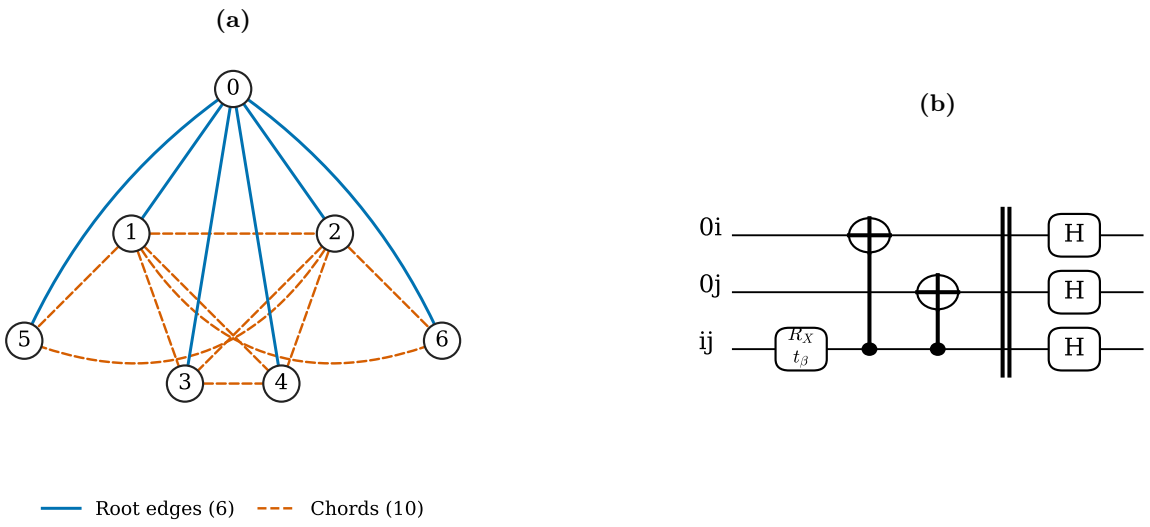



FIG. 7. Triangle-based temperature encoding. Panel (a) shows the seven-vertex graph, with six root edges forming a spanning tree and ten chords. Panel (b) shows the preparation motif for a fundamental triangle. A chord rotation $R_X(t_\beta)$ and two CNOTs prepare $\cos(t_\beta/2)\,|000\rangle - i\sin(t_\beta/2)\,|111\rangle$ on edges $0i, 0j, ij$. The full encoder repeats the rotation and CNOT construction for ten chords with shared root edges, then applies the global Hadamard layer once. The displayed motif is therefore an illustration, rather than an independent three-qubit block repeated with its own Hadamards. Every rotation uses $t_\beta = (\pi/2)\tanh(1.5\beta)$.

probabilities, making full-distribution diagnostics necessary whenever feasible. For example, the unconditional Curie–Weiss baseline has mean profile TVD 0.0281 at $\beta = 0.4$ but full TVD 0.3787. The parity constraint fixes signed mean magnetization but does not establish accurate magnetization magnitude or energy fluctuations.

For the shared Curie–Weiss models, we exclude evaluation temperatures from fitting and configuration selection. However, selection uses exact target information, including a physical-observable criterion, and the temperature range was explored during earlier model design. The results therefore assess generalization within an already studied family and temperature range. Uncertainties across ten independent fits describe training variability, while evaluation on a finite grid does not establish accuracy throughout the temperature interval.

At larger sizes, the sparse-lattice runs demonstrate classical conditional training with a symmetry-preserving parity encoder up to 2048 qubits. Hyperparameter optimization at 64 and 256 qubits yields one configuration used across all six sizes, with substantial reductions in held-out sampled $\mathrm{MMD}^2$. Although these runs use the same encoder construction as the sixteen-qubit Curie–Weiss benchmark, their learned connectivity and separately selected optimization settings differ. Moreover, low sampled $\mathrm{MMD}^2$ does not establish high full-distribution fidelity: the fixed expected mask weights emphasize low-order correlations, and size-dependent kernel bandwidths prevent direct fidelity comparisons across sizes.

Future work should examine richer tractable encoders and optimization strategies that improve conditional accuracy. Broader families of correlated inputs and multidimensional encodings should be assessed alongside state-preparation and estimation costs, with particular attention to calibration of each conditional response. Conditional extensions of spectral Born machines would require corresponding sampling, amplitude-ratio and gradient estimators [32]. Generator selection, initialization and adaptive kernels merit investigation [25–27], including whether gradients from different conditions reinforce or cancel one another. Quantum deployment should assess how noise and circuit partitioning affect the conditional response and imposed symmetries [33, 34]. Practical quantum sampling advantage remains open: our Gaussian-shaped and Curie–Weiss benchmark targets admit efficient classical sampling. Demonstrations would require comparisons with strong classical generative models and sampling-hardness or verification arguments appropriate to the trained conditional circuits.

## VII. CONCLUSIONS

We introduced a conditional framework for IQP Born machines by combining reference-state estimation of Pauli-$Z$ moments and their phase gradients with a conditional MMD objective. Efficient reference-state sampling and amplitude-ratio evaluation yield unbiased classical estimators of the objective and its gradients, enabling joint optimization of one shared IQP unitary across a distribution family. Notably, the encoder need not have IQP form. Furthermore, input-state design can impose exact output symmetries for every condition, indepen-

dently of the learned parameters.

Sixteen-qubit numerical experiments demonstrate the framework's capabilities. Translation of Gaussian-shaped profiles validates the conditioning mechanism, while two-coordinate phase encoding demonstrates generation at center–width combinations absent from training. Both use a shared lattice layer; their exact central binomial law follows from construction. The Curie–Weiss model then tests a larger change in distribution shape: one shared unitary reproduces the temperature-dependent transition from unimodal to bimodal magnetization profiles, while the parity encoder guarantees global spin-flip symmetry. A separate signed-graph construction shows how tractable input encoding imposes six independent switching symmetries without restricting the IQP phases. Finally, scaling runs demonstrate computational feasibility and reductions in held-out sampled $\mathrm{MMD}^2$ up to 2048 qubits. However, these reductions do not establish high full-distribution fidelity. The framework thus supports conditional generation and systematic symmetry control through tractable input-state design, while full-distribution accuracy at larger scales and practical quantum sampling advantage remain open.

## DATA AVAILABILITY

The code and data supporting this study are available on GitHub [28].

## ACKNOWLEDGMENTS

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## Appendix A: Hyperparameters and evaluation protocols

### 1. Common training settings

We train all models using the unbiased estimator of the squared-MMD objective and keep the encoders fixed within each fit. For Gaussian translation and scaling, we initialize the circuit phases independently as $\theta_j \sim \mathcal{N}(0, s_\theta^2)$. The other experiments use the initialization protocols specified in their respective appendices. At each update, we sample one fitting condition and one kernel bandwidth uniformly, then draw a target minibatch with replacement. We draw fresh nonidentity masks and apply the probability correction for excluding the identity mask. Mask budgets count draws, including repeated masks. We draw reference samples independently of the target minibatch and reuse them across masks. Table IV summarizes the translation and scaling settings. Section A 4 a specifies the Curie–Weiss search and completed confirmation and final fitting, with selected hyperparameters in Table VII. Appendix B gives the separate two-coordinate protocol.

For each initialization, we reset the optimizer and draw new circuit phases while reusing the target pools. For the experiments summarized in Table IV, each training run includes all prescribed initializations, and we select the checkpoint with the lowest selection score across them. Reported uncertainty summarizes variation across training runs.

We train the Gaussian translation and scaling models with `Adam` [35], using cosine learning-rate decay. Table IV gives the initial learning rates, final-to-initial learning-rate ratios, and moment coefficients. Both experiments use $\epsilon = 10^{-8}$ and zero weight decay. The sixteen-qubit Curie–Weiss protocol specifies its optimizer settings separately.

TABLE IV. Gaussian translation and final scaling settings per circuit. Mask and reference budgets count draws per update. The Curie–Weiss follow-up uses the separate budgets in Table V. Tuned scaling values are rounded to six significant digits.

| Setting | Gaussian translation | Scaling |
|---|---|---|
| Updates per initialization | 1,024 | 5,000 |
| Initializations per run | 1 | 3 |
| Reported runs | 5 | 3 |
| Training targets per condition | 512 | 10,000 |
| Target minibatch | 128 | 450 |
| Training mask draws | 1,024 | 512 |
| Training reference draws | 1,024 | 1,024 |
| Expected mask weights | $\{1\}$ | $\{2, 4\}$ |
| Checkpoint interval | 32 | 100 |
| Optimizer | `Adam` | `Adam` |
| Learning rate | 0.005 | 0.00189326 |
| Learning-rate schedule | Cosine | Cosine |
| Initialization width $s_\theta$ | 0.25 | 0.0129803 |
| Coefficient $b_1$ | 0.9 | 0.822126 |
| Final-to-initial learning-rate ratio | 0.1 | 0.119851 |
| Coefficient $b_2$ | 0.999 | 0.962419 |

For an expected mask weight $w$, we set the Gaussian bandwidth to

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2, \tag{A1}$$

where $n$ denotes the number of visible qubits and $w$ is the expected mask weight before excluding the identity mask. We weight the bandwidths equally in the objective. For the sixteen-qubit Curie–Weiss follow-up, we tune two of the three expected-weight slots as specified in Table VI. Repeated slots retain their separate sampling probabilities. The upper grid value 7.99 keeps $w < n/2$ and the bandwidth strictly positive.

We report rotation angles in radians and compute sixteen-qubit full and profile TVDs from exact circuit probabilities. Sampled-MMD diagnostics do not enter these calculations.

### 2. Efficient evaluation of the sampled estimator

For the sample values defined in Sec. III C, we compute the pair sums using

$$\sum_{s\neq t} Y_s Y_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{A2}$$

and, since $T_b^2 = 1$,

$$\sum_{b\neq b'} T_b T_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{A3}$$

These identities allow us to evaluate Eq. (33) in $O(S+B)$ arithmetic operations once the sample values are available.

For the sampled-mask estimator in Eq. (42), let $\mathcal{U}$ denote the set of distinct sampled masks and $m_u$ the multiplicity of mask $u$. Since repeated masks use the same batches, we can write

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \sum_{u\in\mathcal{U}} \frac{m_u}{M} \widehat{\Delta_c^2}(u). \tag{A4}$$

This regrouping preserves the sampled objective and its gradient while reducing the number of Pauli moments we evaluate.

### 3. Translation of Gaussian-shaped profiles

Translation uses the $v = 0$, $u = 2c - 1$ slice of the two-coordinate target family and encoder. All forty parameters on the open $4 \times 4$ lattice are independent. At $c = 0.5$, the exact uniform law is fixed for every parameter vector, so no learned transformation contributes at that condition. The encoder also has uniform probabilities at all other conditions, whereas their targets are nonuniform.

Each of five independent fits uses 512 target strings at each of $c \in \{0, 0.25, 0.5, 0.75, 1\}$, for 2560 strings. The initialization draws all forty phases independently from $\mathcal{N}(0, 0.25^2)$. We use 1024 sampled `Adam` updates, initial learning rate 0.005 and cosine decay to one tenth of that value. Each update draws one fitting condition, a minibatch of 128 targets with replacement, 1024 reference samples in chunks of 256 and 1024 nonidentity masks. The bandwidth corresponds to expected mask weight one before nonidentity conditioning. Other optimizer and precision conventions match the two-coordinate protocol.

We evaluate initialization, every 32 updates and the final update, selecting the lowest mean exact full-distribution TVD at $c \in \{0.125, 0.375, 0.625, 0.875\}$. Reporting uses 80 disjoint conditions, $c_j = (j + 1/3)/80$ for $j = 0, \ldots, 79$. We compute exact full and profile TVDs, average over these conditions within each fit, and report the median across fits. Figure 2 averages profiles at four reporting conditions, $c \approx \{0.004, 0.329, 0.667, 0.992\}$, chosen to spread the profiles across the interval. It also shows the median and interquartile range of the reporting mean response. Exact development targets contribute to checkpoint selection. The median terminal-checkpoint mean full TVD is 0.0674.

### 4. Curie–Weiss models

We use the sixteen-qubit target $p(x \mid \beta) \propto \exp[\beta(2|x| - 16)^2/32]$. Every trained circuit contains 16 one-body and 120 two-body IQP phase parameters, without ancillas or parameter tying. The campaign compares shared parity-input, shared product-input, independent parity-input and independent unconditional models. The shared product-input model provides a matched encoder ablation. All current comparisons use the fresh final cohort described below.

For encoder scale $s$, the product input prepares $R_Z(s\beta)R_Y(\pi/2)\,|0\rangle$ on every qubit. The parity input prepares $|+\rangle^{\otimes 16}$ and applies $\exp[-i(s\beta/2)\sum_{i=1}^{15} Z_i Z_{16}]$. This input and the learned $X$-generated IQP unitary preserve global spin-flip invariance. They do not impose full permutation invariance. The unconditional input is $|0\rangle^{\otimes 16}$ and has no encoder-scale parameter.

We fit each shared model at $\beta \in \{0.1, 0.7, 1.3, 1.95\}$. Each independent family contains three circuits fitted directly at $\beta \in \{0.4, 1, 2\}$. One hyperparameter configuration applies jointly to the three circuits in each independent family, while their learned phases remain separate.

Each fitting temperature supplies a fixed pool of 1024 sampled targets per repeat. We reuse these pools across updates and initializations, pair compatible pools and random streams across candidates and matched families within each stage, and use independent streams between screening, confirmation and final fitting. Each update draws a target minibatch of 256, 2048 nonidentity masks and 1024 reference samples. Shared fits sample each fitting temperature with probability 1/4. We sample the three kernel slots with equal probability, retaining duplicate slots when their weights coincide.

TABLE V. Curie–Weiss budgets per circuit. The last column totals optimizer updates across all repeats and initializations. All stages use 2000 updates per initialization and checkpoints at initialization and every 100 updates. Each shared repeat fits one circuit using 4096 target-pool entries; each independent-family repeat fits three circuits using 3072 entries. Thus the independent families use three times the learned phases and optimizer updates per repeat.

| Stage | Repeats | Starts per repeat | Updates per circuit |
|---|---|---|---|
| Screening, per proposal | 5 | 1 | 10,000 |
| Confirmation, per candidate | 5 | 3 | 30,000 |
| Final fitting, per family | 10 | 3 | 60,000 |

### *a. Hyperparameter optimization*

We use full-distribution accuracy as the primary selection criterion, giving the worst-temperature error twice the weight of the temperature-averaged error. When selecting checkpoints and the final hyperparameter configuration, we minimize the physical-observable score among candidates whose primary score lies within 0.01 of the best available value.

For a grid $G$ and nonnegative error curve $e$, define

$$R_G(e) = \frac{2\max_{\beta\in G} e(\beta) + \overline{e}_G}{3}, \tag{A5}$$

where $\overline{e}_G$ is the trapezoidal integral divided by the actual grid span. At a single temperature, both terms equal the point error. We compute each fit's primary score as $J = R_G(\mathrm{TVD})$ using exact probabilities over all $2^{16}$ bitstrings. The physics score is

$$P = \frac{R_G(e_m) + R_G(e_c)}{2}, \quad e_m = \left|\langle|m|\rangle_q - \langle|m|\rangle_p\right|, \tag{A6}$$

with $e_c = |c_{\mathrm{fluc},q} - c_{\mathrm{fluc},p}|/c_\star$ and the fixed target-only normalization $c_\star \approx 0.823110$. Signed mean magnetization and spin-flip defect remain diagnostics and do not enter selection. The fluctuation statistic does not establish thermal-derivative consistency.

Shared checkpoint selection always uses the development grid

$$\begin{aligned} A = {} & \{0.05 + 0.12j : j = 0, \ldots, 15\} \\ & \cup \{1.975\}. \end{aligned} \tag{A7}$$

For each circuit and repeat, we pool every scheduled checkpoint across all available initializations, find the minimum $J$, and choose the smallest $P$ among checkpoints with $J \le J_{\min} + 0.01$. Ties favor smaller $J$, then earlier updates and lower initialization index. The tolerance applies once over the complete pool and does not imply statistical equivalence. Independent circuits use this rule at their directly fitted temperatures.

We assign one `Optuna` study to each of the four model families [36]. Each study comprises 60 independent random proposals followed by 60 adaptive tree-structured Parzen estimator (TPE) proposals, for 120 screening proposals per family. TPE continues the same study using the random-stage observations. We retain the categorical kernel representation chosen during earlier development and do not repeat its pilot. Table VI specifies the fixed search space, and Table VII gives the configurations selected by fresh confirmation.

The random stage provides a probabilistic justification for the initial budget [37]. Under the fixed search distribution, 60 independent proposals have probability $1 - 0.95^{60} \approx 0.954$ of including at least one configuration in its best-performing 5% according to the screening objective. This statement concerns a quantile of the specified search distribution, not proximity to a global optimum or a prescribed accuracy. The subsequent 60 TPE proposals adapt to the observed screening scores; the independent-sampling calculation does not establish their coverage or guarantee final performance.

TABLE VI. Curie–Weiss follow-up search space and initial random sampling distributions. TPE adapts these proposals after the random stage. We fix the optimizer to `Adam`. The encoder scale applies only to conditional families.

| Parameter | Options or range | Initial prior |
|---|---|---|
| Learning rate | $[3 \times 10^{-4}, 10^{-1}]$ | Log-uniform |
| Width $s_\theta$ | $[0.01, 1]$ radians | Log-uniform |
| $b_1$ | $[0.8, 0.95]$ | Uniform |
| $1 - b_2$ | $[0.001, 0.05]$ | Log-uniform |
| Schedule | Constant, cosine | Equal probabilities |
| Encoder scale $s$ | $[0.25, 1.5]$ | Log-uniform |
| Kernel pair $(w_2, w_3)$ | 45 ordered pairs | Uniform categorical |

The kernel slots are $(2, w_2, w_3)$, where $w_2 \le w_3$ and $w_2, w_3 \in \{4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 7.99\}$. Repetitions are allowed.

We fix $\epsilon = 10^{-8}$ and weight decay to zero, with no warmup or gradient clipping. The cosine schedule decreases the learning rate to one tenth of its initial value over 2000 updates. We sample the initial phases from the stated Gaussian without clipping or wrapping. The encoder-scale range keeps $s\beta \le 3 < \pi$ over $\beta \in [0, 2]$; it does not bound the Gaussian circuit phases. We use multivariate TPE with 60 startup observations and 24 expected-improvement candidates, without pruning or queued reference configurations. Numerical failures consume proposal slots and receive a flagged search penalty of 1, which is not a measured TVD. Invalid configurations cannot enter confirmation. All 480 screening proposals were valid, and confirmation and final fitting had no numerical failures.

Screening ranks shared candidates on $A$. For independent families, we combine the three selected temperature cases within each repeat using $R_G$, then average the resulting $J$ and $P$ across the five repeats. Shared candidate scores likewise average repeat scores. TPE receives only the primary score $J$; we retain $P$ and the component curves separately.

The confirmation stage shortlists three or four distinct valid configurations per study from all 120 screening proposals. It includes the best $J$, the best $P$ within $J_{\min} + 0.01$, further candidates ranked by $J$ to reach three, and the global best $P$ if not already included. This yielded three shared-parity candidates and four candidates in each other family. Each candidate receives five fresh repeats with three initializations per circuit. Shared checkpoints still use $A$, but confirmation ranks the selected fits on $A \cup B$, where

$$\begin{aligned} B = \{&0.075, 0.135, 0.23, 0.35, 0.47, 0.59, \\ &0.675, 0.735, 0.83, 0.95, 1.07, 1.19, \\ &1.275, 1.335, 1.43, 1.55, 1.67, 1.79, 1.9, 1.9625\}. \end{aligned} \tag{A8}$$

Independent confirmation uses the direct-fit markers. We select the smallest confirmed $P$ among candidates within 0.01 of the minimum confirmed $J$. Confirmation does not feed further TPE proposals.

After freezing one configuration per family, the final stage fits ten fresh repeats with three initializations per circuit. Checkpoint selection continues to use $A$ for shared models and the fitted temperature for each independent circuit. Reporting retains all ten selected repeat outcomes and computes metrics before their means and sample standard deviations, using the sample denominator 9. It does not select the best repeat. The shared reporting grid contains 101 points: 0, 2 and $(6j - 4)/297$ for $j = 1, \ldots, 99$, with separate evaluations at the common markers. This grid is disjoint from fitting anchors and shared selection grids, and never selects checkpoints or configurations. These finite-grid evaluations do not establish accuracy throughout a continuum.

We reviewed the four studies after their random stages and again after TPE, then completed confirmation and final fitting without changing the objective, grids or selection rule. The common markers are absent from shared fitting and development grids, but their temperature range was explored during earlier development. Independent baselines provide direct-fit marker comparisons, not full-grid estimates. We use double precision for circuit parameters and exact probabilities, while the sampled estimator's matrix products use single precision. Exact probability normalization uses absolute tolerance $10^{-8}$ with no relative tolerance.

### *b. Product-input ablation*

We compare the shared parity model with a shared product input

$$|\phi_{\text{prod}}(\beta; s)\rangle = \bigotimes_{i=1}^{16} R_Z(s\beta) R_Y(\pi/2) |0\rangle_i . \tag{A9}$$

TABLE VII. Configurations selected by fresh confirmation. Independent-family settings apply jointly to three circuits with separately learned phases. Optimizer and unlisted settings follow the frozen protocol. Displayed continuous values are rounded; machine-readable settings retain full precision.

| Parameter | Shared parity | Shared product | Independent parity | Independent unconditional |
|---|---|---|---|---|
| Initial learning rate | 0.00341745 | 0.0300696 | 0.00430287 | 0.0898283 |
| Width $s_\theta$ (radians) | 0.0221053 | 0.0454769 | 0.0300356 | 0.0326103 |
| `Adam` $b_1$ | 0.801879 | 0.826492 | 0.899094 | 0.946166 |
| `Adam` $b_2$ | 0.991964 | 0.998983 | 0.96613 | 0.998814 |
| Schedule | cosine | constant | constant | cosine |
| Encoder scale $s$ | 0.683987 | 0.657269 | 0.582332 | – |
| Expected kernel weights | [2, 4, 7.5] | [2, 4, 5] | [2, 7.5, 7.99] | [2, 5, 5.5] |

TABLE VIII. Matched shared-encoder comparison. Marker columns give exact full-distribution TVD; $J$ and $P$ use the same 101-point reporting grid for both encoders. Values are means ± sample standard deviations over ten fresh final repeats. Lower is better for all errors. The comparison changes input correlations and encoder phase generators as well as enforced spin-flip symmetry.

| Model | $\beta = 0.4$ | $\beta = 1$ | $\beta = 2$ | $J$ | $P$ |
|---|---|---|---|---|---|
| Shared parity | $0.1185 \pm 0.0107$ | $0.1396 \pm 0.0048$ | $0.0832 \pm 0.0103$ | $0.1281 \pm 0.0027$ | $0.0865 \pm 0.0088$ |
| Shared product | $0.1576 \pm 0.0079$ | $0.2242 \pm 0.0113$ | $0.1104 \pm 0.0169$ | $0.2051 \pm 0.0088$ | $0.1393 \pm 0.0107$ |

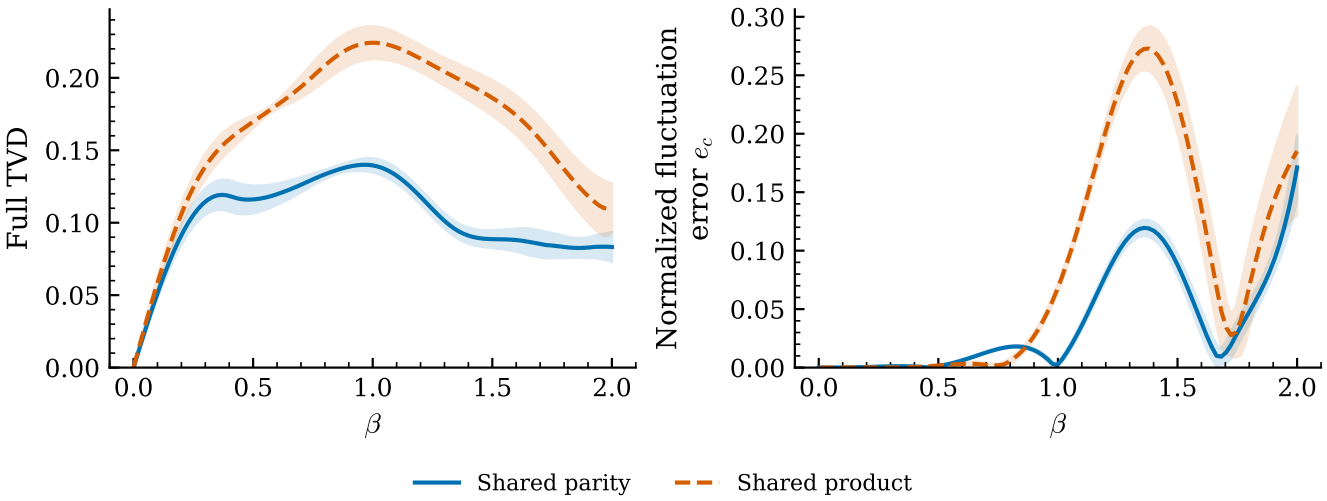



FIG. 8. Matched shared-encoder comparison on the fixed 101-point reporting grid. Curves and shaded bands show means and sample standard deviations over ten fresh final repeats. The left panel shows exact full-distribution TVD; the right shows absolute energy-fluctuation error divided by the fixed target scale $c_\star \approx 0.823110$. Lower is better in both panels. Both models use the same fitting anchors and per-circuit search and training budgets, with separately selected configurations. The uniform output at $\beta = 0$ is fixed by both architectures.

This ablation removes enforced global spin-flip symmetry while retaining temperature encoding in input phases. It also changes the input correlations and encoder phase generators, so a fidelity difference does not isolate symmetry as its sole cause. Both models have uniform computational-basis input probabilities, use 136 independent IQP phases and share the fitting anchors, search ranges and per-circuit budgets. Compatible random streams and target pools are paired. Each encoder has its own HPO study and selected configuration, listed in Table VII. We evaluate both on the same grids using ten fresh final repeats.

Table VIII gives the matched comparison. Shared parity has robust TVD score $J = 0.1281 \pm 0.0027$, compared with $0.2051 \pm 0.0088$ for shared product, a 37.5% reduction in mean score. Its physics score is $P = 0.0865 \pm 0.0088$ versus $0.1393 \pm 0.0107$, a 37.9% reduction. Mean parity TVD is lower at every nonzero point of the reporting grid. Product has inverse-temperature-averaged TVD $0.1662 \pm 0.0063$ and per-repeat maximum TVD $0.2245 \pm 0.0112$, compared with $0.1033 \pm 0.0036$ and $0.1406 \pm 0.0037$ for parity. Observable errors still cross at some temperatures, as Figure 8

TABLE IX. Exact Hamming-weight profile TVD at sixteen qubits. Values are means ± sample standard deviations over ten fresh final repeats, computed within each repeat before aggregation. Lower is better. Shared parity excludes markers from fitting and selection; independent models fit directly at each marker.

| Model | $\beta = 0.4$ | $\beta = 1$ | $\beta = 2$ |
|---|---|---|---|
| Shared parity | $0.0438 \pm 0.0040$ | $0.1203 \pm 0.0046$ | $0.0646 \pm 0.0124$ |
| Independent parity | $0.0149 \pm 0.0065$ | $0.0642 \pm 0.0028$ | $0.1298 \pm 0.0065$ |
| Independent unconditional | $0.0281 \pm 0.0247$ | $0.0833 \pm 0.0158$ | $0.4003 \pm 0.1736$ |

TABLE X. Exact marker mean absolute magnetization errors and normalized energy-fluctuation errors. Values are means ± sample standard deviations over ten fresh repeats; energy errors use $c_\star \approx 0.823110$. Lower is better. Shared markers are excluded from fitting and selection; independent models fit directly.

| Model | Error | $\beta = 0.4$ | $\beta = 1$ | $\beta = 2$ |
|---|---|---|---|---|
| Shared parity | $e_m$ | $0.0167 \pm 0.0017$ | $0.0581 \pm 0.0028$ | $0.0083 \pm 0.0043$ |
| Shared parity | $e_c$ | $0.0009 \pm 0.0002$ | $0.0028 \pm 0.0015$ | $0.1712 \pm 0.0272$ |
| Shared product | $e_m$ | $0.0174 \pm 0.0039$ | $0.0834 \pm 0.0103$ | $0.0130 \pm 0.0066$ |
| Shared product | $e_c$ | $0.0005 \pm 0.0004$ | $0.0683 \pm 0.0034$ | $0.1851 \pm 0.0551$ |
| Independent parity | $e_m$ | $0.0006 \pm 0.0004$ | $0.0023 \pm 0.0018$ | $0.0127 \pm 0.0013$ |
| Independent parity | $e_c$ | $0.0013 \pm 0.0003$ | $0.0475 \pm 0.0042$ | $0.0705 \pm 0.0053$ |
| Independent unconditional | $e_m$ | $0.0022 \pm 0.0028$ | $0.0030 \pm 0.0024$ | $0.2759 \pm 0.1435$ |
| Independent unconditional | $e_c$ | $0.0003 \pm 0.0002$ | $0.0036 \pm 0.0023$ | $2.8935 \pm 1.4896$ |

shows. Lower aggregate $P$ therefore does not imply better agreement for every observable at every condition.

### c. *Profile and full-distribution errors*

Table IX reports profile TVD from the same ten fresh final repeats as the full-distribution comparison. We compute the profile error in each repeat before averaging, rather than the error of the averaged profile. At $\beta = 0.4$, the independent unconditional model has mean profile TVD $0.0281 \pm 0.0247$ but full TVD $0.3787 \pm 0.0034$, exposing errors within Hamming-weight shells. At $\beta = 1$, independent parity has smaller profile TVD than shared parity, $0.0642 \pm 0.0028$ versus $0.1203 \pm 0.0046$, although its full TVD is larger. Profiles and full distributions thus give different rankings, which is why full TVD remains the primary objective.

### d. *Physical observables*

For a distribution $q$, let $\rho_q(k \mid \beta)$ denote its exact Hamming-weight profile and set $m_k = (2k - n)/n$ with $n = 16$. We compute its moments as $\langle m^r \rangle_q = \sum_{k=0}^{n} \rho_q(k \mid \beta) m_k^r$, and its mean absolute magnetization as $\sum_k \rho_q(k \mid \beta)|m_k|$. The target law corresponds to $H = -nm^2/2$ in units $J = k_B = 1$. Excluding self-interactions would add only a constant to $H$ and would not change the energy variance. Equation (54) therefore becomes

$$c_{\mathrm{fluc},q}(\beta) = \frac{\beta^2 n}{4} \left( \langle m^4 \rangle_q - \langle m^2 \rangle_q^2 \right). \tag{A10}$$

The observables depend only on the shell profile and do not test uniformity within shells. For the Gibbs target, differentiating the mean energy per spin with respect to $T = \beta^{-1}$ gives this fluctuation expression. We do not assume that the learned family satisfies the same derivative identity.

The final observable sweep evaluates ten fresh selected circuits per shared family on the fixed reporting grid, with parameters frozen after checkpoint selection on $A$. The independent baselines appear only at their fitted markers. We compute each observable, its absolute target error and the nonlinear fluctuation statistic within each repeat before taking means and sample standard deviations. Table X reports $e_m$ and $e_c$ at the common markers for all four families. On these same markers, $P$ is $0.0914 \pm 0.0102$ for shared parity, $0.1130 \pm 0.0223$ for shared product, $0.0363 \pm 0.0018$ for independent parity and $1.2223 \pm 0.6289$ for independent unconditional models. These common-marker scores differ from the shared full-grid scores in Table VIII. The large unconditional fluctuation error at $\beta = 2$ accompanies the unstable distributional fits and remains in the reported cohort. The uniform output at $\beta = 0$ has binomial shell law $\rho_0(k) = 2^{-n}\binom{n}{k}$ and is exact for both shared encoders independently of training.

### 5. Scaling

We use the parity encoder in Eq. (51), generalized to $n$ visible qubits, without ancillas. Its selected scale is $s = 1.18430$, and it preserves global spin-flip symmetry at every size. The learned unitary uses the open rectangular lattice architectures in Table XI. At size $n$, the target distribution is $p(x \mid \beta) \propto \exp[\beta(2|x| - n)^2/(2n)]$.

We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, using fixed pools of 10,000 target samples per condition. We draw a separate validation pool of 2000 targets per training condition. Held-out temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$ enter reporting only and do not select checkpoints or hyperparameters.

TABLE XI. Open rectangular lattice architectures for the scaling experiments. We assign one trainable phase to each qubit and one to each horizontal or vertical nearest-neighbor pair.

| Visible qubits | Lattice | Trainable parameters |
|---|---|---|
| 64 | $8 \times 8$ | 176 |
| 128 | $8 \times 16$ | 360 |
| 256 | $16 \times 16$ | 736 |
| 512 | $16 \times 32$ | 1,488 |
| 1,024 | $32 \times 32$ | 3,008 |
| 2,048 | $32 \times 64$ | 6,048 |

Checkpoint selection uses four mask-and-reference banks evaluated on the same validation pools. Each bank uses 512 nonidentity mask draws per condition and bandwidth, and 1024 reference draws per condition, shared across bandwidths. We reuse each bank's random numbers across checkpoints and pair compatible streams across candidates within each search stage. Let $\overline{M}_n^2(\boldsymbol{\theta}; \beta)$ denote the sampled validation $\mathrm{MMD}^2$ averaged over the four banks and two bandwidths. We select checkpoints using

$$V_n(\boldsymbol{\theta}) = \frac{1}{4} \sum_{\beta \in \{0.6, 0.8, 1.2, 1.4\}} \frac{\overline{M}_n^2(\boldsymbol{\theta}; \beta)}{d_{n,\beta}}. \tag{A11}$$

Each fixed scale $d_{n,\beta}$ is the median independently evaluated training-temperature $\mathrm{MMD}^2$ across three earlier baseline runs, floored at $10^{-3}$. These candidate-independent scales balance the temperatures' contributions to selection. Training continues to sample conditions and bandwidths uniformly, and reporting uses unnormalized $\mathrm{MMD}^2$. We retain negative unbiased estimates.

We optimize the six hyperparameters in Table XII. The first 16 proposals comprise a baseline configuration, a proposal from earlier sixteen-qubit parity-model optimization, and 14 random proposals. The remaining 32 proposals use a tree-structured Parzen estimator.

TABLE XII. Scaling hyperparameter search space and initial random sampling distributions.

| Hyperparameter | Range | Initial random sampling |
|---|---|---|
| Initial learning rate | $[10^{-4}, 0.05]$ | Log-uniform |
| Initialization width $s_\theta$ | $[0.005, 0.3]$ | Log-uniform |
| Encoder scale $s$ | $[0.25, 1.5]$ | Uniform |
| $b_1$ | $[0.8, 0.95]$ | Uniform |
| $b_2$ | $[0.95, 0.999]$ | Uniform |
| Final-to-initial learning-rate ratio | $[0.05, 1]$ | Log-uniform |

Screening evaluates each of the 48 proposals at 64 and 256 qubits, with one initialization and 2000 updates per size. Each proposal's score averages its minimum checkpoint-selection score across the two sizes. We confirm the three leading proposals using three fresh runs per size, one initialization per run, and 5000 updates. We also evaluate the baseline under the same confirmation budgets. The lowest mean confirmation score determines the configuration used at all six sizes.

Final fits use the settings in Table IV, with three runs per size and three initializations of 5000 updates per run. The runs use separate target pools and stochastic training streams. However, their initialization sets overlap, giving five distinct starting parameter vectors per system size. Within each run, the initializations reuse target pools and stochastic training streams, while resetting the optimizer and drawing new phases. We evaluate initialization and every 100 updates, then select the lowest-scoring checkpoint across all three initializations.

For each run, we report the selected initialization's trajectory at update 0 and every 500 updates through update 5000. At each reporting update, we average five independent $\mathrm{MMD}^2$ estimates over the held-out temperatures and bandwidths. Each estimate uses 2000 target draws per condition, 512 mask draws per condition and bandwidth, and 1024 reference draws per condition. Reporting samples are independent of training and selection, reused across updates, and share reference samples across bandwidths. Figure 6 shows the median and interquartile range across the three run-level averages. These summaries combine variation from training and sampled evaluation.

## Appendix B: Two-coordinate encoder and evaluation protocol

### 1. Fitting, development and reporting conditions

We use the binomial-centered target family in Eq. (47) and the phase encoder in Eq. (49). Grid training tests interpolation, while axis training tests new combinations inside and outside the joint fitting hull. Fitting conditions drive sampled MMD updates, development conditions select configurations and checkpoints, and reporting conditions assess the selected models.

Grid fitting uses $\{-1, 0, 1\}^2$ and development uses

$$\{-1, -0.5, 0, 0.5, 1\} \times \{-0.75, -0.25, 0.25, 0.75\}.$$

The reporting grid is

$$u \in \{-1, -0.83, -0.37, 0.19, 0.61, 1\},$$
$$v \in \{-1, -0.79, -0.31, 0.23, 0.67, 1\}.$$

Excluding its four fitted corners leaves thirty-two pairs.

Axis fitting uses $(u, 0)$ at $u \in \{-1, -0.37, 0, 0.37, 1\}$ and $(0, v)$ at $v \in \{-1, -0.31, 0.31, 1\}$. Development uses $(u, 0)$ at $u \in \{-0.73, -0.17, 0.17, 0.73\}$ and $(0, v)$ at $v \in \{-0.69, -0.13, 0.13, 0.69\}$. Reporting uses sixteen mixed pairs,

$$\{-1, -0.37, 0.37, 1\} \times \{-1, -0.31, 0.31, 1\}.$$

Four satisfy $|u| + |v| \leq 1$ and lie inside the fitting hull; twelve lie outside. Each individual coordinate value appears in fitting, but no mixed reporting pair appears in fitting or development. These sets were explored before the present campaign, so they test the fitted models within an already studied domain.

### 2. Sampled training and exact development selection

Each fit uses a fixed pool of 1024 target bitstrings at each of nine fitting conditions. An update draws one condition and one bandwidth uniformly, 256 target strings with replacement, 1024 fresh reference samples and 1024 nonidentity masks. Expected mask weights before conditioning on nonidentity are $\{2, 4\}$, with bandwidth $\sigma = (-2\ln(1-2w/16))^{-1/2}$ and the identity-mask correction. Conditional on the pool, sampled MMD targets the empirical distribution. We use `Adam` with $b_1 = 0.9$, $b_2 = 0.999$, $\epsilon = 10^{-8}$ and zero weight decay. Cosine decay reduces the learning rate to one tenth of its initial value. Estimator accumulation uses double precision, matrix products use single precision, and reference batches use chunks of 256.

For each geometry, we screen four settings for 512 updates. The sixteen one-body parameters are independent draws from $\mathcal{N}(m, 0.1^2)$, with $m \in \{0, \pi/3\}$; the twenty-four pair parameters are independent draws from $\mathcal{N}(0, 0.1^2)$. Initial learning rates are 0.01 or 0.03. We retain the two lowest development scores, run two 2000-update confirmations per candidate, and choose the lower mean confirmation score. Both geometries select $m = \pi/3$ and learning rate 0.01. We freeze both configurations before reporting evaluation, then run five independent final fits per geometry. Screening, confirmation and final fitting total twenty-six fits.

Checkpoints minimize mean exact full-distribution TVD on development conditions. We evaluate initialization, every 64 screening updates or 100 updates otherwise, and the final update. Pools and stochastic streams are paired across settings and geometries at each repeat, while final repeats are independent of configuration selection. This procedure uses exact development targets in addition to finite fitting pools; it is not a finite-data-only pipeline. The phase coefficients and family design also reflect preceding exploratory work. All ten final fits select a checkpoint after at least 1000 updates. The nonzero initialization branch gives lower development scores in this search, so the campaign does not establish robustness across initialization regimes.

Independent checks compare reference samples, amplitude ratios, circuit probabilities and sampled losses and gradients with dense or public circuit calculations. At four qubits, 512 independent sampled estimates agree with an exact empirical-pool MMD calculation within 0.67 standard errors for the loss and 1.43 standard errors for every gradient coordinate. A sixteen-qubit check at three condition pairs agrees with direct circuit probabilities to below $6 \times 10^{-18}$. These checks test implementation agreement at the stated settings, rather than bounding estimator tails throughout the domain.

### 3. Reference access and circuit preparation

The input amplitude for a bitstring of weight $k$ is

$$a_{u,v}(k) = 2^{-8} \exp\left[\frac{iu(k-8)}{2} - \frac{iv(k-8)^2}{40}\right]. \tag{B1}$$

For a reference bitstring of weight $j$, grouping Hadamard signs by the overlap of occupied positions gives

$$\psi_{u,v}(j) = 2^{-8} \sum_{k=0}^{16} a_{u,v}(k) \mathcal{K}_k(j), \tag{B2}$$

$$\mathcal{K}_k(j) = \sum_{\ell=0}^{k} (-1)^\ell \binom{j}{\ell} \binom{16-j}{k-\ell}, \tag{B3}$$

with invalid binomial coefficients set to zero. We tabulate seventeen reference amplitudes, draw $j$ with probability $\binom{16}{j}|\psi_{u,v}(j)|^2$, then draw a uniform weight-$j$ bitstring. The ratio for a sample $z$ and mask $b$ uses table entries at $|z|$ and $|z \oplus b|$, dividing only on sampled support. At the center, the reference is exactly $|0\rangle^{\otimes 16}$. For general $n$, these operations require polynomial work. However, efficient access does not ensure that the chosen sample budgets resolve small losses or gradient components.

Preparation starts with sixteen Hadamard gates, followed by sixteen $R_Z(u/2)$ rotations and 120 pairwise $ZZ$ rotations. Since

$$J_z^2 = \frac{n}{4} I + \frac{1}{2} \sum_{i<j} Z_i Z_j,$$

applying $\mathrm{IsingZZ}_{ij}(v/40) = e^{-ivZ_iZ_j/80}$ to every pair implements the quadratic phase up to a global phase. No ancillas or target amplitude preparation are required. The learned lattice layer adds sixteen one-body and twenty-four nearest-neighbor two-body phases with independent parameters. These are high-level gate counts; the experiment does not assess compiled depth or deployment noise.

### 4. Full-distribution evaluation and response diagnostics

The reference remains permutation symmetric, but the independently parameterized lattice generally breaks this symmetry. We therefore compute all $2^{16}$ output probabilities for evaluation and use full-distribution TVD as the primary metric. Profile TVD aggregates probabilities by Hamming weight and may hide errors within shells. Dense evaluation is a small-system diagnostic, distinct from classical sampled training.

We average reporting errors over conditions within each fit and report means and sample standard deviations across five fits. The two geometries have different reporting sets, so their columns do not define a paired accuracy ranking. Terminal-checkpoint mean full TVDs are $0.0489 \pm 0.0137$ for axis training and $0.0412 \pm 0.0087$ for grid training. We retain these outcomes to distinguish final-update performance from development-selected performance.

After selection, we evaluate all 1681 pairs on $u, v \in \{-1, -0.95, \ldots, 0.95, 1\}$. Figure 3 shows the pointwise arithmetic mean of full TVDs, rather than TVD of the mixture, without smoothing. This sweep includes fitting and development conditions and is diagnostic.

We measure endpoint response ratios by dividing each model's mean change between $u = -1$ and $u = 1$, or its width change between $v = -1$ and $v = 1$, by the corresponding exact target change. Averaged across five fits, mean-displacement ratios at $v = (-1, 0, 1)$ are $(0.966, 1.000, 1.043)$ for axis training and $(0.964, 0.998, 1.042)$ for grid training. Width-change ratios at $u = (-1, 0, 1)$ are $(0.917, 0.829, 0.917)$ and $(0.898, 0.814, 0.898)$, respectively. Individual fits recover between 0.507 and 1.073 of the target width change. None reverses an adjacent width step on the tested

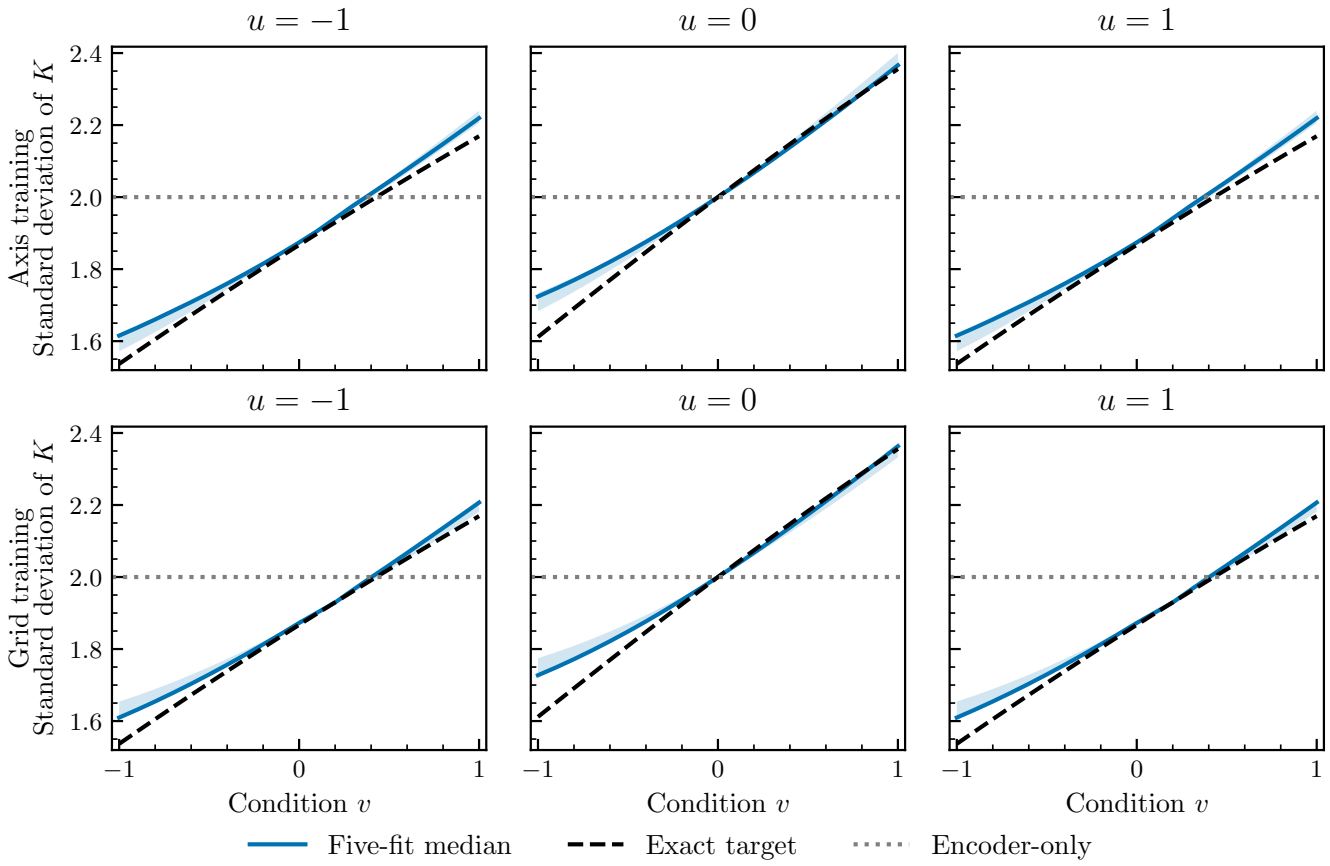



FIG. 9. Width response with independently parameterized lattice layers. Columns fix $u = -1, 0, 1$; rows show axis and grid fitting. Blue curves and shading show the pointwise median and interquartile range across five selected fits. Dashed black curves show actual target standard deviations, and dotted gray lines show the encoder's constant binomial width. These 21-point sweeps are post-selection diagnostics and include previously examined conditions.

sweeps. These summaries describe response magnitude on the tested slices, not perfect calibration throughout the domain. In particular, agreement at the origin holds for every parameter vector and is excluded from the separate reporting averages.

## Appendix C: Binary-linear conditional encoders

Injectivity of $C$ makes the states $|Cu\rangle$ mutually orthogonal. The factor normalization in Eq. (55) therefore gives $\langle\psi_c|\psi_c\rangle = 1$ and the independent-bit sampling law stated in Section V.

Precompute a binary left inverse $L$, satisfying $LC = I_k$. For a sampled configuration $z$, recover its latent coordinates as $u = Lz$. For a mask $a$, compute $d = La$ and test whether $Cd = a$. If this equality fails, translation by $a$ takes every reference configuration outside $\operatorname{im} C$. Otherwise, translation changes $u$ to $u \oplus d$, giving the product ratio in Eq. (57). After polynomial preprocessing, sampling and ratio evaluation require $O(nk + n)$ elementary operations, assuming bounded arithmetic cost for each factor.

Extend the columns of $C$ to an invertible binary matrix $M$. Prepare the $k$ single-qubit factor states and initialize the remaining visible qubits in $|0\rangle$. A reversible linear circuit implementing $M$ maps $|u, 0\rangle$ to $|Cu\rangle$. Standard binary elimination gives an implementation using $O(n^2)$ CNOT gates with unrestricted connectivity, without additional ancillary qubits [38]. Applying the final Hadamards prepares $|\phi(c)\rangle$. Efficient deployment also requires efficient preparation of the single-qubit factors.

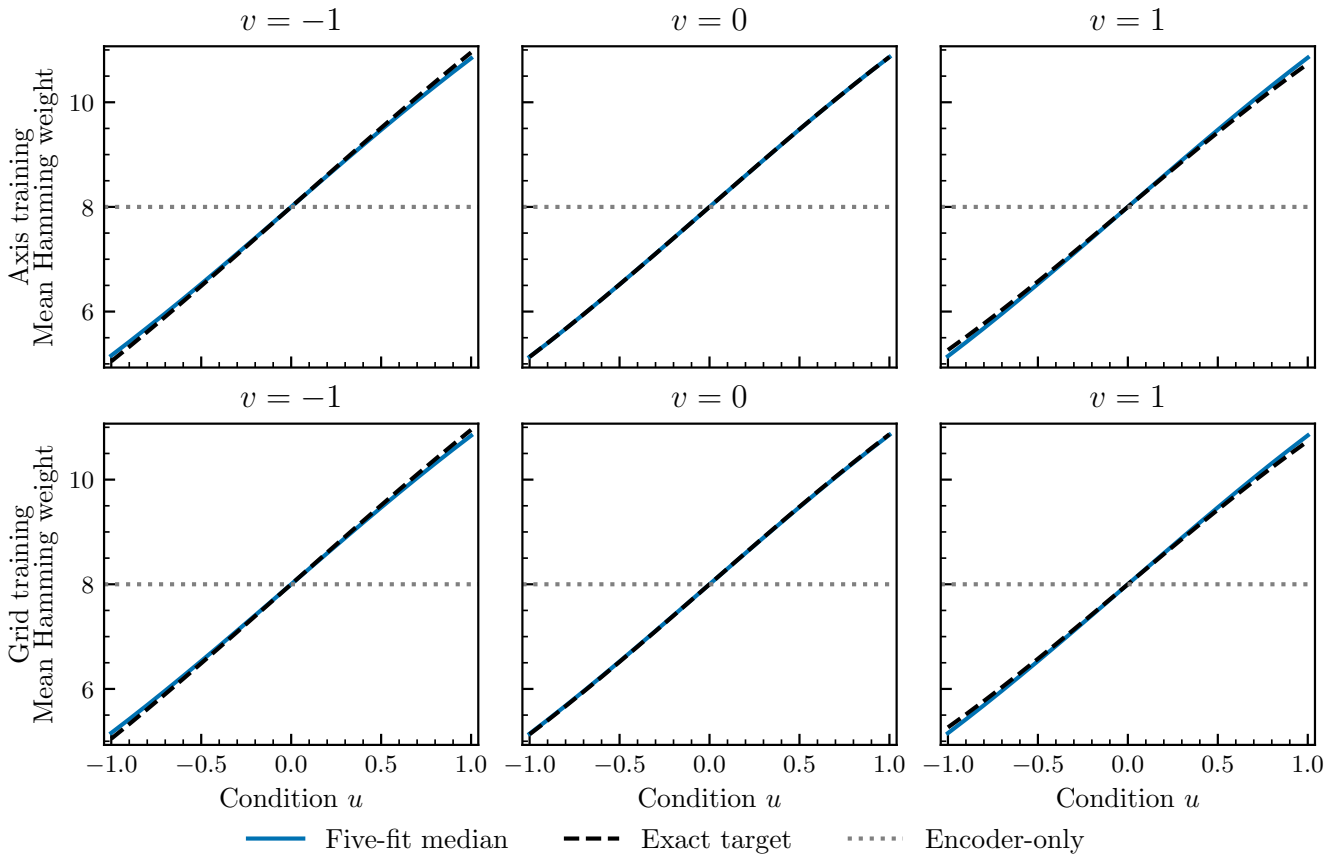



FIG. 10. Mean response for the same selected fits. Columns fix $v = -1, 0, 1$; rows show axis and grid fitting. Curves and shading follow Figure 9. The target mean depends on both coordinates, while the encoder mean is eight at every condition.

## Appendix D: Sixteen-qubit conditional switching construction

### 1. Graph geometry, cycle reference and symmetry

We specialize the binary-linear encoder framework of Section V to the graph's cycle space. This construction enforces the six independent vertex-switching symmetries.

We order the sixteen edges lexicographically as

$$(01, 02, 03, 04, 05, 06, 12, 13, 14, 15, 16, 23, 24, 25, 26, 34). \tag{D1}$$

The chord set is $\mathcal{C} = \{12, 13, 14, 15, 16, 23, 24, 25, 26, 34\}$. Each chord and its two root edges form a fundamental triangle. We represent these ten triangles using a binary matrix $B \in \{0,1\}^{10\times 16}$. Each row contains ones on its triangle's three edges and zeros elsewhere. The chord columns form the identity matrix, so $B$ has rank ten over $\mathbb{F}_2$. The connected graph has cycle rank $16 - 7 + 1 = 10$, so these rows form a complete cycle basis. In the notation of Section V, the encoding matrix is $C = B^{\mathsf{T}}$. The six independent vertex switches generate the set $G$ of 64 switching masks. Each vertex switch flips either zero or two edges of every triangle, which gives $Bg = 0$ for every $g \in G$.

We denote the ten chord bits by $r \in \{0,1\}^{10}$ and write $|r|$ for their number of ones. Before the final Hadamard layer, we prepare the reference state

$$|\psi_\beta\rangle = \sum_{r\in\{0,1\}^{10}} a_\beta(r)\,|rB\rangle\,, \tag{D2}$$

$$a_\beta(r) = c_\beta^{10-|r|}(-is_\beta)^{|r|}, \tag{D3}$$

where $c_\beta = \cos(t_\beta/2)$, $s_\beta = \sin(t_\beta/2)$, and $t_\beta = (\pi/2)\tanh(1.5\beta)$. The product $rB$, evaluated modulo two, specifies

the sixteen edge bits obtained by combining the triangles selected by $r$. Applying the Hadamard layer gives the encoded input $|\phi_\beta\rangle = H^{\otimes 16} |\psi_\beta\rangle$.

To sample the reference distribution classically, we draw the ten chord bits independently, assigning each a value of one with probability $s_\beta^2$, and return $rB$. We recover $r$ directly from the chord bits because the corresponding columns of $B$ form the identity matrix.

To test whether a bit-flip mask $m$ belongs to the cycle space, we read its ten chord entries into $u$ and check whether $uB = m$, with all operations evaluated modulo two. If this equality holds, applying the mask maps $rB$ to $(r \oplus u)B$. For nonzero $\beta$, we evaluate the amplitude ratio as

$$\frac{a_\beta(r \oplus u)}{a_\beta(r)} = \left[ -i \tan\left( \frac{t_\beta}{2} \right) \right]^{|r \oplus u| - |r|} . \tag{D4}$$

If $m$ lies outside the cycle space, the flipped configuration has zero reference amplitude, so the ratio is zero. At $\beta = 0$, the reference state is $|0\rangle^{\otimes 16}$: the ratio equals one for $m = 0$ and zero otherwise. We evaluate ratios only at sampled configurations with nonzero reference amplitude. These operations provide classical sampling and amplitude-ratio access without enumerating the full sixteen-qubit distribution.

Because $Bg = 0$, every supported configuration $rB$ has even parity on the edges selected by the switching mask $g$. The corresponding Pauli-$Z$ string therefore satisfies $Z^g |\psi_\beta\rangle = |\psi_\beta\rangle$. The Hadamard layer converts this relation into $X^g |\phi_\beta\rangle = |\phi_\beta\rangle$. Every IQP gate generated by a Pauli-$X$ string commutes with these switching operators, so the learned circuit preserves switching symmetry for arbitrary parameter values.